Structure, Properties and Performance of Solder Joints in Terrestrial vs. Reduced-Gravity Environments

Final Report

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Project Overview

This research provides quantitative correlations between solidification and flow characteristics, processing conditions, microstructure, and porosity for solder joints produced in terrestrial vs. microgravity conditions (onboard the International Space Station (ISS)). Furthermore, this project focuses on analyzing solder joint properties under extreme conditions of elevated and cryogenic temperatures in space. Soldering is the predominant method used to create electrical/mechanical joints and mechanical/pressure joints and is typically a rapid process in which the solder material is molten for only a few seconds. Solder joint porosity is a common but undesirable feature naturally arising from the use of fluxes and is more insidious in soldering joints formed under microgravity conditions. Under the presence of gravity, voids and bubbles are removed from the solder joints due to presence of buoyancy force and vigorous liquid solder mixing. In the *absence* of gravity, fluid motion is greatly reduced, driven mainly by solidification volume change, thermally induced density gradients, and Marangoni effects. With slower mixing, voids and bubbles are not swept away and become entrapped in the interior of the solder joint upon solidification. This entrapped porosity dramatically degrades the thermal and electrical properties of the solder and severely reduces the mechanical integrity of the joint. Therefore, as part of the ongoing endeavors to establish an electronics and mechanical joint repair capability for extended space missions, this project seeks to advance the current understanding of fundamental mechanisms, phenomenology, and process conditions that govern the integrity and performance of solder joints produced in terrestrial vs. reduced gravity environments. Additionally, solder joints in the space environment are subjected to large temperature variations such as – 120 ºC to +25 ºC for the Mars missions, and – 157 ºC to +121 ºC outside ISS. However, as deep space exploration missions become longer and more distant, the solder and other metal joints' current reliability is inadequate for the harsh conditions they encounter.

This project focused on the following objectives to address the reliability of solder joints in space: 1) Characterizing the effects of the lack of Earth's natural convective flow and buoyancy on solder-joint formation onboard the ISS using the 40Pb-60Sn solders from the In-Space Soldering Investigation (ISSI), 2) Investigating the effects of elevated and cryogenic temperature conditions in space on solder properties. Considering a range of potential applications and materials, three solder alloys were investigated, including the ISSI lead-based (40Pb-60Sn) solders, off-eutectic 50Pb-50Sn solders, and lead-free (95.5Sn-3.8Ag-0.7Cu) solders, which had recently shown promise for high-performance joint applications due to their good thermal and electrical conductivities, as well as excellent corrosion and fatigue resistance.

Our team conducted the microstructural characterization of four microgravity-processed solders from the ISSI study, one ISSI terrestrial solder, and a third set of freshly made terrestrial 40wt%Pb-60wt%Sn solder to demonstrate the effects of the absence of buoyancy and natural convection on void formation in microgravity. The effect of void-induced Marangoni convection on the solder microstructure was demonstrated using the microgravity wire-feed solder. Additionally, the micromechanical response of solders processed in terrestrial and microgravity environments was investigated under extreme elevated and cryogenic temperature conditions (ranging from –157ºC to +121ºC, as experienced outside the ISS). Under cryogenic conditions, the study investigated the impact of the β-to-α phase transformation in Sn below 13ºC, including associated volume changes, internal stress development, and the ductile-to-brittle transition observed in various Sn-based

solder alloys. At room temperature, the effects of accelerated aging on the solders' microstructure and mechanical properties were explored.

Major Goals/Objectives

The major focus of this project was to elucidate the fundamental mechanisms, phenomenology, and process conditions that govern the integrity and performance of solder joints produced in terrestrial vs. reduced gravity environments, such as the microgravity conditions onboard the International Space Station (ISS). The primary objectives included advancing the current qualitative understanding of porosity formation and persistence in solder joints into the realm of alloy/process-specific quantitative description and prediction and evaluating the influence of elevated and cryogenic temperatures on the microstructural and mechanical properties of solder joints, with emphasis on Sn-based alloys.

Leveraging solder samples from the In-Space Soldering Investigation (ISSI) and other non-ISSI compositions, the project employed a combination of advanced experimental techniques and computational modeling. These approaches were designed to characterize the impacts of void-induced microstructural variations, extreme thermal cycling, and phase transformations, addressing critical reliability challenges for soldering in space environments. The project aimed to address critical scientific gaps related to solder joint reliability in the unique microgravity and thermal conditions of space, with the ultimate goal of providing quantitative insights for designing more robust soldering techniques. While this proposal focused on understanding the fundamental phenomena, it also laid a solid scientific foundation for future advancements in space fabrication and repair technologies. This work contributes directly to NASA’s goals of improving in-space manufacturing and repair capabilities while establishing a strong foundation for future innovations in soldering processes tailored to the harsh conditions of deep-space exploration.

Accomplishments under Goals

The research program achieved several significant accomplishments in advancing the understanding of solder behavior in terrestrial vs. microgravity environments, with a focus on enhancing their reliability for space applications. This study analyzed near-eutectic 40Pb-60Sn solder samples from the In-Space Soldering Investigation (ISSI), conducted in microgravity aboard the International Space Station (ISS) and under terrestrial conditions at NASA Marshall. The ISS experiments were performed by taking pieces of silver-coated copper wire and bending and twisting them into multiple test coupon configurations. The selected lengths of 40wt%Pb-60wt%Sn wire were cut and 1) wrapped around the test coupons and melted using the soldering iron – Wire wrap solder and 2) fed on the similar test coupons that were heated by the soldering iron – Wire feed solder (Fig.1). The ground experiments involved conductively melting 10 cm lengths of 40Pb-60Sn (3.3 wt% rosin flux) solder onto a loop turned into the silver coated copper wire. For more details about the ISSI experiments, refer to: Background: In-Space Soldering Investigation (ISSI 2003-04) section in this report.

Additionally, near-eutectic 40Pb-60Sn, off-eutectic 50Pb-50Sn, and 95.5Sn-3.8Ag-0.7Cu terrestrial solder samples (named ISU terrestrial) were prepared and examined in Dr. Pathak's research lab at Iowa State University, Ames. For a detailed list of samples and tasks, refer to the Outline of Executive Summary of this report. Key accomplishments include:

Accomplishments from the study on near-eutectic 40Pb-60Sn solder (microgravity and terrestrial):

Samples analyzed:

- ISSI microgravity 40Pb-60Sn wire warp solder (Microgravity test 2 sample 4, 5 and 10)
- ISSI microgravity 40Pb-60Sn wire feed solder (Microgravity test 4 sample 16 – MT4S16)
- ISSI terrestrial 40Pb-60Sn wire feed solder
- ISU Terrestrial 40Pb-60Sn wire wrap and wire feed solder (prepared in Dr. Pathak's lab)
- ISU Terrestrial 40Pb-60Sn Reflow Spreading solder (prepared in Dr. Pathak's lab)

- Demonstrated the effect of natural convection and buoyancy on void size in 40Pb-60Sn solder formed under microgravity vs. terrestrial conditions using non-destructive 3-D tomography. Results demonstrated that the occurrence of larger voids (as large as 1 mm in diameter) only happens in microgravity conditions.
- Developed a quantitative framework for describing and predicting porosity formation and persistence in 40Pb-60Sn solder under microgravity vs. terrestrial conditions, addressing the role of reduced buoyancy-driven convection and flux volatilization. ISSI microgravity wire wrap 40Pb-60Sn solder had ~13× more voids than ISSI terrestrial 40Pb-60Sn solder, while ISSI microgravity wire feed 40Pb-60Sn solder had 4× more voids than ISSI wire wrap 40Pb-60Sn solder.
- Studied the effect of Marangoni convection in microgravity 40Pb-60Sn solder under the presence of a large free surface (large void, greater than 1 mm) on solder microstructure (primary dendrite and small void accumulation). The average Pb-particle size near the hot site was 6.1 ± 5 µm and near the cold site (expected to be at a lower temperature than the hot site) was 4.8 ± 5 µm which was caused by the differential cooling rate in the colder and hot sites and accumulation of Pb-rich dendrite in the hot region.
- Quantified the effect of microstructure, influenced by void-induced Marangoni convection, on microgravity solder properties. We demonstrated that Inhomogeneity (accumulation of Pb-rich

dendrites at the solder joint) resulting from Marangoni flow caused lower strength (hardness ~ 0.11 GPa) of the solder at the hot site than at the colder site (hardness ~ 0.15 GPa).

- The aging study on terrestrial 40Pb-60Sn solder showed a hardness decrease from 0.19 GPa on day 1 to 0.16 GPa on day 483, accompanied by Pb particle growth from 1.28 μm to 2.45 μm over the same period.
- Investigated the β-Sn to α-Sn phase transformation below 13°C on pure Sn single crystal and terrestrial 40Pb-60Sn solder using high-throughput spherical nanoindentation, elucidating the effect of associated volume changes and internal stresses on mechanical performance in cryogenic environments.
- Investigated the effects of ductile to brittle transition on terrestrial 40Pb-60Sn solder properties via in-situ SEM micropillar compressions. Plastic strain at instability under compression in 40Pb-60Sn decreased from $\varepsilon = 0.109$ at RT to $\varepsilon = 0.06$ below -85°C, suggesting a ductile-to-brittle transition near this temperature.

Accomplishments from the study of terrestrial off-eutectic 50Pb-50Sn solder:

Sample analyzed:

  - ISU Terrestrial 50Pb-50Sn solder sample 1 to 3 (prepared in Dr. Pathak's lab)
- Aging study on terrestrial 50Pb-50Sn demonstrated that the hardness dropped from 0.18 to 0.14 GPa, and Pb particle size grew from 0.45 to 0.8 μm over 89 days.

Accomplishments from the study on terrestrial 95.5Sn-3.8Ag-0.7Cu solder:

Sample analyzed:

  - ISU Terrestrial 95.5Sn-3.8Ag-0.7Cu solder sample 8 (prepared in Dr. Pathak's lab)

- In-Situ SEM micro-pillar compression on ISU terrestrial 95.5Sn-3.8Ag-0.7Cu solder demonstrated that plastic strain at instability under compression dropped from $\varepsilon = 0.06$ at RT to $\varepsilon = 0.03$ at -85°C and lower temperatures, signaling a ductile-to-brittle transition around this temperature.
- The deformation behavior was correlated with microstructure across all temperatures. Group 3 (eutectic mixture of Ag_3Sn + Sn) demonstrated a sharp decrease in the plastic strain at instability as the temperature drops below room temperature (RT), probably caused by the high fraction of the Ag_3Sn intermetallic phase that significantly influences the mechanical response. At cryogenic temperatures, Ag_3Sn is much stiffer and more brittle than the Sn-rich phase. As temperature decreases, deformation becomes increasingly localized at phase boundaries, reducing plasticity and accelerating the ductile-to-brittle transition.
- In contrast, Group 2a (eutectic Ag_3Sn + Sn + Sn-rich dendrite) showed a gradual decrease in plastic strain with decreasing temperature, as its balanced mix of Sn-rich and eutectic phases allowed for some residual plasticity, unlike the sudden drop observed in Group 3. Group 2b (Cu_5Sn_6 + group 2a) exhibited the lowest plastic strain across all temperatures due to the presence of Cu_6Sn_5 intermetallic compounds, which increased stiffness and restricted plastic deformation, leading to early instability.

Objectives and Outcomes

Year 1:

Sample analyzed:

- ISSI M 40Pb-60Sn Test 2 Sample 4
- ISSI M 40Pb-60Sn Test 4 Sample 16
- ISSI 40Pb-60Sn Terrestrial Sample 2
- ISSI 40Pb-60Sn Terrestrial Sample 3
- ISU 40Pb-60Sn Terrestrial Sample 1
- ISU 40Pb-60Sn Terrestrial RSS Sample 10

The first year of this project was focused on studying the near-eutectic 40Pb-60Sn (terrestrial and microgravity) and off-eutectic 50Pb-50Sn (terrestrial) solder compositions. This study focused on understanding the role of reduced buoyancy forces and natural convection in void formation and microstructural inhomogeneity in 40Pb-60Sn solder under microgravity conditions, alongside the effects of room temperature aging on ISSI microgravity 40Pb-60Sn, terrestrial 40Pb-60Sn and terrestrial 50Pb-50Sn solder properties. Additionally, the study aimed to quantify the influence of Marangoni flow and thermal convection on void dynamics and phase segregation in ISSI microgravity 40Pb-60Sn solder. The study demonstrated that microgravity conditions significantly increased void fraction (ISSI microgravity wire wrap solder had ~13× more voids than terrestrial solder) and size in solder joints due to the absence of buoyancy-driven convection, with Marangoni flow contributing to void and Pb-rich dendrite accumulation in high-temperature regions. Mechanical testing revealed ductile-to-brittle transitions in ISU terrestrial 40Pb-60 Sn solder under cryogenic conditions at temperatures at and below -85 ^{o}C, while aging studies highlighted grain coarsening (Pb particle size =1.28 μm on day 1 to 2.45 μm on day 483) and hardness reduction (0.19 GPa on day 1 to 0.15 GPa on day 483) in 40Pb-60Sn solder alloys.

Please see the year one report for further details of this section.

Year 2:

Sample analyzed:

ISSI M 40Pb-60Sn Test 2 Sample 10

ISSI M 40Pb-60Sn Test 4 Sample 16

ISU 40Pb-60Sn Terrestrial Sample 3

ISU 50Pb-50Sn Terrestrial Samples 1, 2, and 3

ISU 95.5Sn-3.8Ag-0.7Cu Terrestrial Sample 8

The second year of this project focused on 1) quantifying the effects of void-induced Marangoni convection and thermal gradients on ISSI microgravity 40Pb-60Sn solder strength and 2) evaluating the influence of elevated and cryogenic temperatures on Sn-based terrestrial solders (40Pb-60Sn, 50Pb-50Sn and 95.5Sn-3.8Ag-0.7Cu) properties, including ductile-to-brittle transition and phase transformations.

The study revealed that void-induced Marangoni convection in microgravity significantly reduces solder hardness near high-temperature regions due to the localized accumulation of softer Pb-rich phases, highlighting the critical need to develop soldering processes that mitigate microstructural

inconsistencies to ensure the structural integrity of solder joints in space environments. Cryogenic temperature studies on ISU terrestrial 40Pb-60Sn solder using high throughput spherical nanoindentation stress-strain analysis revealed that 40Pb-60Sn solder exhibits increased yield strength from 0.18 GPa (0.05 offset yield strength) at RT to 0.32 GPa at -120°C, while pure Sn shows unusual softening at -120°C, likely due to the β-to-α phase transformation, emphasizing the need for alloy-specific reliability assessments under extreme conditions. The micro-pillar compression experiments revealed a significant ductile-to-brittle transition in ISU terrestrial 40Pb-60Sn solder below -85°C, characterized by decreased plastic strain at instability from 0.109 at RT to 0.062 at -85°C and deformation concentrated along Pb-rich and Sn-rich phase boundaries. Micropillar compression experiments on ISU terrestrial 95.5Sn-3.8Ag-0.7Cu solder at cryogenic temperatures revealed a significant increase in stress at instability which changed from 78 MPa at RT to 153 MPa at -145°C, with a noticeable saturation in instability stress below -85°C and a reduction in the plastic strain at instability (from 0.06 at RT to 0.03 at -85°C), primarily attributed to a ductile-to-brittle transition. Deformation mechanisms in 95.5Sn-3.8Ag-0.7Cu varied with temperature and percentage of various phases in the different pillars, with deformation predominantly progressing along Sn-rich dendrites at -40°C, shifting to eutectic and intermetallic interfaces below -85°C, highlighting the critical role of phase boundaries in cryogenic mechanical behavior.

Dissemination

Summary of Technical Dissemination:

Research undertaken and supported by this grant was completed at Iowa State University Ames by graduate and undergraduate researchers. Scientific products were finalized and disseminated through professional society technical presentations, invited academic, and technical seminars, and student posters at several academic and society events. While most technical presentations were given by the PIs, graduate researchers supported by this project had many opportunities for presentation and publishing.

At the time of this final report's submission, four peer-reviewed publications are in the final stages of preparation for submission.

Further, outcomes from research activities in this program were included as 1 invited talk and 12 conference talks, including graduate student Manish Kumar's one PhD dissertation and one more invited talk to be given after the completion of this grant.

Products:

2025

- Kumar M, Napolitano R, Pathak S, "Reliability of Terrestrial vs. Microgravity Solders under Extreme Elevated and Cryo Temperatures for Deep Space Exploration" , The Minerals, Metals and Materials Society (TMS), , Las Vagas, NV March 23–27, **2025**.

2024

- Kumar M, Napolitano R, Pathak S, "Reliability of Terrestrial vs. Microgravity Solders under Extreme Elevated and Cryo Temperatures for Deep Space Exploration" Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), San Juan, Puerto Rico December 3-7, 2024.
- Kumar M, Napolitano R, Pathak S, "Reliability of Terrestrial vs. Microgravity Solders under Extreme Elevated and Cryo Temperatures for Deep Space Exploration" International Space Station Research & Development Conference (ISSRDC), Boston, Massachusetts, July 29-August 1, 2024.
- **Invited:** Kumar M, Napolitano R, Pathak S, "Structure and Properties of Terrestrial vs. Microgravity Solders under Extreme Conditions of Elevated and Cryo Temperatures," Virtual NASA PSI User Meeting, March 2024.
- Kumar M, Napolitano R, Pathak S, "Structure and Properties of Terrestrial vs. Microgravity Solders under Extreme Conditions of Elevated and Cryo Temperatures," The Minerals, Metals and Materials Society (TMS), Orlando, Florida, March 3-7, 2024.
- Kumar M, Napolitano R, Pathak S, "Structure and Properties of Terrestrial vs. Microgravity Solders under Extreme Conditions of Elevated and Cryo Temperatures," Center for Advanced Non-Ferrous Structural Alloys IRB Review Meeting, Iowa State University, Ames, Iowa, October 22-23, 2024.

2023

- Kumar M, Napolitano R, Pathak S, "Structure and Properties of Terrestrial vs. Microgravity Solders under Extreme Conditions of Elevated and Cryo Temperatures," Annual Meeting of

the American Society for Gravitational and Space Research (ASGSR), Washington, D.C, November 14-18, 2023.

- Hellyer S, Ogden C, Kumar M, Napolitano R, Pathak S, "Studying the Effects of Aging on the Structure and Properties of Off-Eutectic 50wt%Pb-50wt%Sn Solder Joints for In-Space Applications," Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Washington, D.C, November 14-18, 2023.
- Kumar M, Napolitano R, Pathak S, "Structure and Properties of the Terrestrial vs. Microgravity Solders under Extreme Conditions of Elevated and Cryo Temperatures," International Space Station Research & Development Conference (ISSRDC), Seattle, Washington, July 31-August 3, 2023.
- Hellyer S, Ogden C, Kumar M, Napolitano R, Pathak S, "Studying the Effects of Aging on the Structure and Properties of Off-Eutectic 50wt%Pb-50wt%Sn Solder Joints for In-Space Applications," National Conference on Undergraduate Research **(NCUR),** Eau Claire, Wisconsin, April 13-15, 2023.
- Kumar M, Jacob K, Napolitano R, Pathak S, "Structure and Properties of Pb-Sn Solders Produced in Terrestrial vs. Microgravity Environments," Center for Advanced Non-Ferrous Structural Alloys IRB Review Meeting, Colorado School of Mines, Golden, Colorado, September 27-28, 2023.
- Kumar M, Napolitano R, Pathak S, "Structure and Properties of Solder Joints Produced in Terrestrial and Microgravity Conditions," The Minerals, Metals and Materials Society (TMS), San Diego, California March 19-23, 2023.
- Kamp J, Madison N, Kumar M, Pathak S, "Structure and Properties of Pb-Sn and Pb-Free 95.5Sn-3.8Ag-0.7Cu Solders Produced in Terrestrial Environment," ISU Symposium on Undergraduate Research and Creative Expression, Ames, Iowa, April 20, 2023.

Scientific Manuscripts under preparation

- **Kumar, M.**, Azizi, G., Napolitano, R., Zaeem, M., Pathak, S., "Effect of Aging on Structure and Properties of the 40Pb-60Sn Solders for In-Space Applications."
- **Kumar, M.**, Napolitano, R., Pathak, S., "Structure and Properties of Solder Joints Produced in Terrestrial vs. Microgravity Conditions."
- **Kumar, M.**, Hellyer, S., Ogden, C., Kamp, J., Napolitano, M., Pathak, S., "Effect of Aging on Structure and Properties of the 50Pb-50Sn Solders for In-Space Applications."
- **Kumar, M.**, Napolitano, R., Pathak, S., "Solder Joint Reliability under Extremes of Elevated and Cryogenic Temperatures Experienced during Deep Space Exploration."

Honors and Awards

2024

- December 2024: Graduate student Manish Kumar's research was prominently featured in NASA's 2025 Science Calendar for the month of November.
  Hear Dr. Fox's thoughts on our research during the ASGSR conference:
  https://iastate.box.com/s/gov6zufb17rvnleoua075k8n4nf1y4dw
  Link to download the high-resolution NASA Science calendar:

https://science.nasa.gov/multimedia/2025-nasa-science-planning-guide/?utm_source=cal&utm_medium=print&utm_id=2025guide



- March 2024: Undergraduate researcher Soren Hellyer was awarded the TMS Acta Materialia Scholarship-2024
- February 2024: Manish Kumar was awarded the Brown Graduate Fellowship Award ($10,000) – Recognized by the Office of the Vice President for Research at Iowa State University for exceptional achievements and contributions to space science.
- January 2024: TMS student travel award ($300) - to help defray the costs of attendance at the Mineral, Metals and Materials Society annual meeting (TMS-2024).

2023

- November 2023: Manish Kumar was awarded ASGSR Student Travel Award ($500) to help defray the costs of attendance at the American Society for Gravitational and Space Research (ASGSR-2023) conference.
- August 2023: Manish Kumar was awarded Wayne G. Basler Scholarship ($5000) for 2023 in Materials Science and Engineering at Iowa State University
- June 2023: Sigma Xi Grants ($1000) in Aid of Research 2023 - Sigma Xi's Committee on Grants-in-Aid of Research has approved Manish's research grant proposal titled" Solder Joints Under Extreme Conditions of Temperature and High Strain-rate for In-Space Applications," submitted for the March 15, 2023 application cycle. Among the 97 recipients of the Sigma Xi award this year, only seven were granted in the field of Engineering, and Manish was one of those seven recipients. You can read the press release about the recipients: https://www.sigmaxi.org/programs/grants-in-aid-of-research/grant-recipients
- April 2023: Noah Madison was awarded the Outstanding First-Year Honors Mentor Program (FHMP) Scholar Award – 2023

- March 2023: Joshua Kamp and Noah Madison were awarded First-Year Honors Mentor Program Grant-2023
- February 2023: Dr. Pathak was awarded the 2023 Iowa NASA EPSCoR Partnership Development Travel Grant ($7500) to initiate a new research collaboration on soldering under extreme temperature conditions with NASA International Space Station (ISS) implementation partners and scientists at NASA Marshall
- January 2023: Soren Hellyer and Caleb M Ogden received the NASA – Iowa Space Grant Consortium Award ($5000). This award was intended to provide experiences to undergraduates that promote skills for STEM careers through hands-on activities (Project mentors: Manish Kumar, Dr. Sid Pathak)

Training Opportunity

Graduate researchers who were involved in this project (PhD student at ISU - Mr. Manish Kumar) got the opportunity to do the following training and experiences:

- Received training at the Sensitive Instrument Facility (SIF) at Ames Laboratory, including SEM/FIB, EBSD, and in-situ SEM Hysitron PI-85 Picoindenter to perform both in-situ nanoindentation and micropillar compression.
- Visited Bruker Inc., Eden Prairie, MN, to perform In-Situ SEM Nanoindentation experiments at cryo temperature and received training on TI-980 Triboindenter.
- Received training on Micro-EDM machine to fabricate the micro-pillars and micro-tensile geometries.
- Traveled to the Advanced Photon Source in Argonne National Laboratory for Non-destructive 3-D tomography experiments.
- Traveled to the University of Wisconsin, Madison to fabricate the tensile gripper using the Plasma FIB.

Undergraduate students (Soren Hellyer, Caleb Ogden, Noah Madison, Joshua Kamp) were trained on the automatic mechanical polisher (Tegramin-25), microstructural characterization tool (Optical microscope and SEM), and nano-mechanical testing tool (TI-950 nanoindenter).

Participants

Name	Role
Sid Pathak	PI
Ralph Napolitano	Co-PI
Manish Kumar	Graduate Research Assistant
Soren Hellyer (Graduating in Spring 2025)	Undergraduate Research Assistant
Caleb Ogden (Graduating in Spring 2025)	Undergraduate Research Assistant
Joshua Kamp	Undergraduate Research Assistant
Noah Madison	Undergraduate Research Assistant
Justine Cowley	Undergraduate Research Assistant

Outline of Executive Summary

This project was conducted under six major tasks. Each task was composed of integrated experimental (at Iowa State University, ISU, and Advanced Photon Source, APS) and modeling (at Iowa State University, ISU). The study utilized both microgravity and terrestrial solder samples from ISSI experiments, along with additional samples prepared under terrestrial conditions in Dr. Pathak's lab at Iowa State University (samples named ISU terrestrial). The sample list includes:

Table 1: List of solder samples analyzed in the NASA PSI soldering project, including samples from the In-Space Soldering Investigation (ISSI) and those prepared in Dr. Pathak's lab.

	Sample name	**Sample preparation method**
Microgravity	ISSI M 40Pb-60Sn Test 2 Sample 4	Wire Wrap Solder (Parallel Cu wire)
	ISSI M 40Pb-60Sn Test 2 Sample 5	Wire Wrap Solder (Parallel Cu wire)
	ISSI M 40Pb-60Sn Test 2 Sample 10	Small Loop Wire Wrap Solder (Small Loop Cu wire)
	ISSI M 40Pb-60Sn Test 4 Sample 16	Wire Feed Solder (Small Loop Cu wire)
Terrestrial	ISSI Terrestrial 40Pb-60Sn Sample 2	Wire Feed Solder (Small Loop Cu wire)
	ISSI Terrestrial 40Pb-60Sn Sample 3	Wire Feed Solder (Parallel Cu wire)
	ISU Terrestrial 40Pb-60Sn Sample 1	Wire Wrap Solder (Parallel Cu wire)
	ISU Terrestrial 40Pb-60Sn Sample 2	Wire Feed Solder (Parallel Cu wire)
	ISU Terrestrial 40Pb-60Sn Sample 10	Reflow Spreading Solder (Cu Foil)
	ISU Terrestrial 50Pb-50Sn Samples 1, 2, and 3	Reflow Spreading Solder (Cu Foil)
	ISU Terrestrial 95.5Sn-3.8Ag-0.7Cu Sample 8	Reflow Spreading Solder (Cu Foil)

A breakdown of these three thrust areas is shown below in table 1.

Table 2: Overview of tasks conducted in the NASA PSI soldering project and the corresponding solder samples analyzed in each task.

Task	**Task Area**	**Sample Used**	**Year**
Task A	3-D microstructural characterization	ISU Terrestrial 40Pb-60Sn Sample 2	Year 1
		ISSI Terrestrial 40Pb-60Sn Sample 3	
		ISSI M 40Pb-60Sn Test 2 Sample 5	
Task B	Sample mounting and mechanical polishing	Samples used for microstructural characterization	Year 1
Task C	Microstructural characterization	ISSI M 40Pb-60Sn Test 2 Sample 4	Year 1 and Year 2
		ISSI M 40Pb-60Sn Test 2 Sample 10	
		ISSI M 40Pb-60Sn Test 4 Sample 16	
		ISSI Terrestrial 40Pb-60Sn Sample 2	

		ISSI Terrestrial 40Pb-60Sn Sample 3	
		ISU Terrestrial 40Pb-60Sn Sample 1	
Task D	Micro-Mechanical Testing Strategies for Solders at Elevated and Cryo Temperatures	ISU Terrestrial 40Pb-60Sn Sample 2	Year 1 and Year 2
		ISU Terrestrial 40Pb-60Sn Sample 10	
		ISU Terrestrial 95.5Sn-3.8Ag-0.7Cu Sample 8	
Task E	Aging effects	ISU Terrestrial 40Pb-60Sn Sample 2	Year 1 and Year 2
		ISU Terrestrial 50Pb-50Sn Samples 1, 2, and 3	

This final report focuses on Tasks C, D, and E. Please see the year 1 report (Link) for a detailed description of results from Tasks A, B, and C.

Project Motivation

Metal joining (welding, brazing, and soldering) in the space environment provides a solution towards sustainable, flexible missions through on-demand fabrication and repair of sustained damage to the spacecraft, satellites, and habitat structures [1, 2]. Metal joining in space is subjected to: a) reduced gravity, b) reduced pressure, and c) a large temperature gradient than terrestrial metal joining. Once the metal joints are produced, they experience extreme temperature variations such as – 120ºC to +25ºC for Mars missions, – 230ºC to +120ºC for the missions to Moon, and – 157ºC to +121ºC outside the ISS. Understanding the metal joining parameters in microgravity and joint integrity during welding, brazing, and soldering under extreme temperatures in space is critical for the safety and success of NASA's space exploration missions to and beyond low Earth orbit. In this proposed research, we primarily focus on (microgravity vs. terrestrial) solder joints because: ● Solder alloys are low melting alloys due to which they experience the largest change in microstructural and mechanical properties under extreme temperatures. ● Solder alloys are an important class of materials for in-space manufacturing and metal joining, as they serve as mechanical and electrical connectors between components and substrates in electronic packaging and are used for mechanical/pressure joints.

Solder joints in electronic components are routinely subjected to large temperature variations in the space environment. To compensate, long-duration missions such as Mars rover missions have traditionally kept their electronics in a heated enclosure known as a "warm electronics box" (WEB) so that they do not get exposed to temperature ranges beyond –40ºC to +40ºC [3]. This approach of protecting electronics from extreme temperatures reduces the spacecraft payload ratio, increases launch costs, and fails to protect the electronics, such as solar arrays, exposed outside. Therefore, future missions are planning to use a distributed, non-heated architecture for the rover and spacecraft electronics, where the components will be placed near the systems they interact with rather than in a heated enclosure [4]. This new approach relies on low-cost, commercially available, and space-qualified electronic packaging assemblies. With this approach, electronic components, without requiring heater power or WEB, must be capable of operating under the extreme temperature experienced in these longer and more distant deep space exploration missions [5, 6].

Addressing the reliability of solder joints under these multi-extreme environments is critical for the success of future deep space exploration missions. Specifically, the solder joints used in onboard electronics must be able to withstand elevated and cryogenic temperatures and large variations in temperature. For example, the ISS orbits Earth 16 times in 24 hours and experiences 16 sunsets and sunrises, due to which anything on the outer surface of the ISS experiences extreme cryo and elevated temperature 16 times in a day.

Scientific Questions and Problem Statement

The first year of this project focused on understanding the factors contributing to increased void accumulation in solder joints prepared under microgravity conditions, specifically the higher number and larger size of voids. For a detailed discussion of these findings, refer to the Year 1 report. Building on this foundation, the second year of the study shifted focus toward assessing the reliability of these solder joints in space environments.

Solder joints are crucial in electronic packaging, serving as mechanical and electrical connectors between components and substrates. A single failure in these solder joints can lead to the malfunction of the integrated circuit or even the entire device. Notably, solder joint failures are the

primary cause of 70% of onboard electronic failures attributed to package failure [7-9]. Therefore, to address the solder joint's reliability, this project focuses on the following question: *How do extreme temperatures, both elevated and cryogenic, experienced by the solder joint in space affect its properties?*

Under Cryogenic Temperature:

- Sn-based solders undergo a ductile to brittle transition (DBT) at the cryo temperature leading to brittle crack formation, which can cause the catastrophic failure of the solder joints.
- Sn-based solders generally occur a β-Sn to α-Sn phase transition below 13ºC where β-structure (Body-centered tetragonal) transforms to α-structure (Diamond cubic). This phase transformation causes a 26% increase in volume that generates internal stresses, due to which the reliability of the Sn-based solder joints is compromised for space applications. Although there is existing literature on this subject [5, 6, 10], a comprehensive understanding of how these phase transformations influence the solder properties remains elusive.

The β-Sn to α-Sn transformation is influenced by several factors, including grain size, the presence of specific solutes, and seeding with compatible materials. Coarser grain structures, characterized by fewer heterogeneous nucleation sites such as grain boundaries, corners, and edges, exhibit slower transformation rates. Elements like Pb, Bi, and Sb, which have good solubility in Sn, hinder the transformation by pinning dislocations within the Sn matrix, making lattice expansion energetically unfavorable. In contrast, seeding with materials like α-Sn, InSb, and CdTe, which possess similar lattice parameters, accelerates nucleation and promotes the transformation process [5, 6, 11-15]. Additionally, the finer grains tend to lower the DBTT, while coarser grains raise it. Finer grains provide more grain boundaries, which act as obstacles to crack propagation. Cracks have to change direction repeatedly as they traverse the grain boundaries, requiring more energy and delaying brittle fracture. This helps maintain ductile behavior to lower temperatures, reducing the DBTT [16]. The solder microstructure in a microgravity environment generally exhibits coarser features compared to those formed under terrestrial conditions. This coarsening is primarily due to the altered dynamics of heat and mass transfer in the absence of gravity-driven natural convection. Without buoyancy forces, fluid motion is significantly reduced, and heat transfer relies predominantly on diffusion, which is much slower. Consequently, the slower cooling rates allow grains and phases to grow larger, resulting in a coarser microstructure. As a result, the kinetics of the β-Sn to α-Sn transformation are expected to slow down in microgravity, while the ductile-to-brittle transition temperature (DBTT) is likely to increase. To address these challenges, it is critical to optimize soldering process parameters, alloy composition, and microstructure to inhibit the β-Sn to α-Sn transformation at cryogenic

Table 3. Selected solder compositions for investigating the effects of elevated and cryogenic temperatures on solder properties.

Solder Composition	Notes
40Pb-60Sn	Near Eutectic, It has better tin phase transformation resistance and low-temperature performance compared to lead-free solders.
50Pb-50Sn	Off eutectic solder with higher melting range than 40Pb-60Sn. It has better tin phase transformation resistance and low-temperature performance compared to lead-free solders. Additionally, it has lower surface tension than 40Pb-60Sn, which is beneficial to reduce void amount in microgravity conditions.
95.5Sn-3.8Ag-0.7Cu	A reliable Lead-free alternative for leaded solders. Sn-Ag-Cu (SAC) solder offers excellent mechanical properties, including high tensile strength and shear resistance

temperatures and reduce the DBTT, ensuring the reliability of solder joints in extreme environments.

Under Elevated Temperatures: Because of the high homologous temperature ($RT/T_m = 0.68$) of the solder alloys, classical elemental diffusion described by Fick's law initiates grain coarsening[17], which can affect the performance and reliability of these solder joints for space applications. Such an aging process is further accelerated when subjected to elevated temperatures in space.

As deep space exploration missions become longer and more distant, the solder joints' current reliability is inadequate for the harsh conditions they encounter [5, 6], and needs to be further improved. This research aims to fill these knowledge gaps, enhancing the reliability of solders for space missions. Additionally, under thermal cycling, thermal expansion mismatch of different components in solder joints causes the stress gradient, which can lead to intermetallic growth followed by brittle failure [7].

Considering a range of potential applications and materials, 3 solder alloys were investigated (Table 3), including the ISSI lead-based (near eutectic 40Pb-60Sn) solders, off-eutectic 50Pb-50Sn solders, as well as lead-free (95.5Sn-3.8Ag-0.8Cu) solders, which have recently shown promise for high-performance joint applications due to their good thermal, electrical conductivities and excellent corrosion, fatigue resistance [18-20].

Research Impact/Earth Benefits

The ongoing project is unique in that it combines both experimental and computational research on solder samples from the microgravity and ground based ISSI experiments, along with a comparative study on fresh solder samples. This project has the following research impact: (i) Our project objectives are a timely alignment with current NASA initiatives related to metals joining in reduced gravity environments. The three solder alloys for this work were chosen after extensive discussions with our NASA collaborators, and these alloy systems are currently being considered at NASA *as possible candidates for use in the NASA Mars Rover sample return missions*. (ii) We also anticipate that the results from this project will inform experiment design responsive to the (future) NASA proposal calls such as the NASA Artemis program, which seeks to land the first woman and next man on the Moon. (iii) The insights gained from this work can be extended to study effects of fluid interfaces with surface tension variations that are also common for many spaceflight-enabling technologies (e.g., boiling, heat transfer, combustion, welding, brazing, or soldering). We envision that this project will contribute towards advanced understanding of multiple processes that typically rely on gravity-driven separations in the terrestrial environment, such as the removal of gas bubbles from a liquid, which must be accomplished by other means in microgravity environments. (iv) the developed gravity-level-specific protocols (GLSPs) for several configurations can be used by future scientists and flight personnel for studying metal joining techniques such as welding, brazing, or soldering under reduced gravity.

Additionally, this project studies the reliability of the solder joints under the extremely elevated and cryo temperatures experienced during the moon and Mars rover missions, which will have a high impact on the success of future deep space exploration. NASA is currently planning future landings to the Moon's south pole, which is filled with deep, dark craters and has been sunless for billions of years. In these areas, the temperature can drop to -203 degrees C, which makes remote observation from Earth difficult and presents problems for operating sensitive equipment in these regions. For example, India's Chandrayaan-3 mission landed Vikram lander and Pragyan rover

near the moon's southern pole to study the mineral composition of the lunar surface and collect data to look for signs of frozen water (essential for rocket fuel). After spending a little under two weeks exploring the promising area, the rover had been "set into sleep mode" with its scientific instruments turned off. However, according to ISRO, it could not survive the harsh lunar night, and its hardware probably succumbed to the moon's freezing conditions. More recently, JAXA's Smart Lander for Investigating Moon (SLIM) landed near the Shioli crater, south of the lunar equator, and as of today SLIM has entered a two-week dormancy period during the long lunar night. According to JAXA, SLIM is not designed for the harsh lunar night. For the SLIM probe to wake up again during the next lunar day, however, its electronic components must survive nighttime temperatures of around -130 degrees C. Reliability of electronic components and metal joints under these harsh extremes of temperature in deep space is an urgent issue that needs more investigation.

Therefore, the insight gained from this research will ensure the reliability of the electronic components used in rover missions by studying the properties of different solder compositions under extremely elevated and cryo temperatures experienced during space missions and assist in the development of new solder alloys that can withstand harsh temperatures. Furthermore, this research will directly support NASA's initiative of using a distributed, non-heated architecture for the rover and spacecraft electronics that does not require heater power or a "warm electronics box" (WEB).

Background: In-Space Soldering Investigation (ISSI 2003-04)

This project utilizes the solder samples from the *ISSI experiments* performed in terrestrial and microgravity conditions (onboard ISS). ISSI experiments aboard the ISS present a unique opportunity to improve our fundamental understanding of the soldering phenomena in a reduced gravity environment. The experimental procedure consisted of 86 soldering experiments conducted on the ISS in 2003-04, and an equivalent set of ground-based experiments using identical equipment and supplies. The ISS experiments were performed by taking pieces of silver-coated copper wire and bending and twisting them into multiple test coupon configurations. The selected lengths of 40wt%Pb-60wt%Sn wire were cut and 1) wrapped around the test coupons and melted using the soldering iron 2) fed on the similar test coupons that were heated by the soldering iron (Fig.1). The ground experiments involved conductively melting 10 cm lengths of 40Pb-60Sn (3.3 wt% rosin flux) solder onto a loop turned into the silver coated copper wire as well as straight silver coated copper wire.

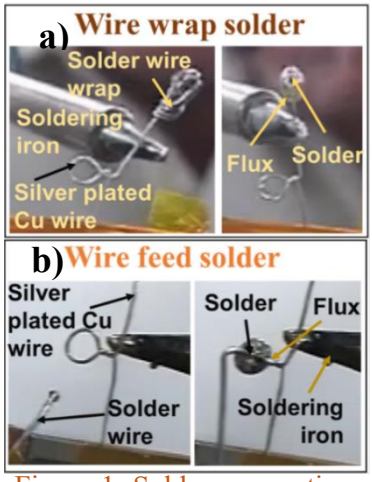


Figure 1: Solder preparation aboard the ISS a) Wire wrap solder b) Wire feed solder.

Results from the ISSI experiments performed aboard the ISS have shown that soldering in microgravity is expected to be considerably different than their ground-based counterparts due to Earth's natural convective flow and buoyancy effects being minimized in microgravity during melting and solidification [21-23]. On Earth, tests showed the solder melting, sagging, dropping, or running down the test wire due to the buoyancy (density) effect (Fig. 2a). In comparison, aboard the ISS, the molten solder was observed to form an equilibrium "football" shape (Fig. 2b). The changes in geometric shape are believed to be due to the dominance of surface tension in microgravity [24-26]. Shortly after equilibrium, the accrued flux was seen spinning around the

sample's surface. As the solder started melting, the internal core of flux was released. When the solder was nearly melted, the released flux formed a drop and migrated to the bottom of the solder, which was closest to the heat source (soldering iron). Shortly after, the flux droplet started to spin around the molten solder. The rotation rate increased steadily and then reached a plateau. This spinning continued until the heat source was removed, after which the flux drop slowed down and eventually stopped. The variation in flux rotation in three samples was probably because of different temperature gradients imposed by different soldering iron tip positions on Cu wire. The heat is initially transmitted through highly conductive Cu wire, which melts the solder. The solder edge closest to the heat source is the hottest, while the opposite edge is likely the coldest, which causes the temperature gradient across the solder surface. Therefore, flux rotation on the sample surface gave indirect evidence of gravity-independent thermocapillary flow induced by temperature gradients across the molten solder ball in µg conditions.

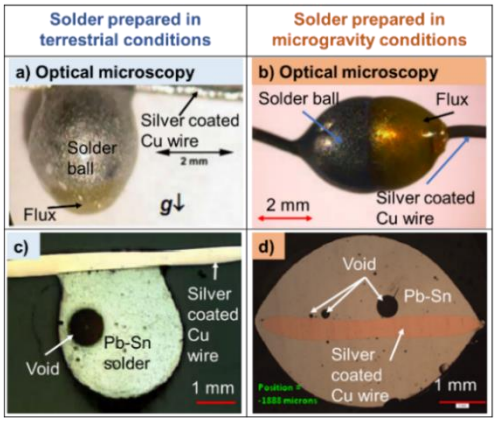


Figure 2: Structure of the terrestrial (a, c) vs. microgravity (b, d) solder using optical microscopy. Change in the geometric shape of the solder is due dominating surface tension in microgravity.

Also, under typical soldering conditions, rapid reflow temperatures results in flux volatilization and bubble formation. In the presence of gravity, liquid solders tend to mix vigorously during the reflow period due to natural convection. This is beneficial, as any voids or bubbles that form due to evaporated fluxes are quickly swept to free surfaces and are thus removed from the solder joint. However, the situation is worse in the reduced gravity environment of space (~1×10^{-5}g) where, due to minimized buoyancy, bubbles have less chance to escape, thus complicating any in-space fabrication and repair operations. The ISSI data has demonstrated that in the *absence* of gravity, fluid motion is greatly reduced, driven mainly by solidification volume change, thermally induced density gradients, and Marangoni effects. Therefore, due to the lack of buoyancy force and slower mixing in the absence of natural convection, voids, and bubbles are not swept away and become entrapped in the interior of the solder joint upon solidification in microgravity conditions (Fig. 2c vs. d). During the process of soldering, the liquid solder temperature near the substrate, which is in contact with the heater, is higher than that far away from the substrate/heater. Therefore, a distinct surface tension gradient will form with low surface tension in the high-temperature region and high surface tension in the low-temperature region. The effect of such a temperature gradient is predominant in the absence of the buoyancy force and natural convective flow [27]. Then the Marangoni force will tend to pull the voids towards the high-temperature regions such as the substrate.

Summary of Year 1

Task B: Sample Mounting and Mechanical Polishing:

All solder specimens were prepared by embedding them in epoxy and then polishing the surface to a very fine finish. A very fine surface finish ensures that the samples are ready for both structural characterization as well as nano-mechanical testing, both of which are very sensitive to surface damage.

Task C: Microstructural Characterization:

Microstructure of ISSI microgravity vs. ISSI terrestrial solder vs. ISU terrestrial:

- Sample analyzed: ISSI microgravity 40Pb-60Sn wire warp solder
- ISSI microgravity 40Pb-60Sn wire feed solder
- ISSI terrestrial 40Pb-60Sn wire feed solder
- ISU Terrestrial 40Pb-60Sn wire wrap and wire feed solder (prepared in Dr. Pathak's lab)

In terrestrial conditions, heat and mass transfer are affected by gravity-induced buoyancy convection. However, in the *absence* of the Earth's natural convective flow, fluid motion is greatly reduced, driven mainly by solidification volume change, thermally induced density gradients, and Marangoni convection. Thermocapillary or Marangoni convection is a surface tension-driven flow in the presence of the temperature gradient [28, 29], causing the liquid to flow from the low surface tension region to the high surface tension region.

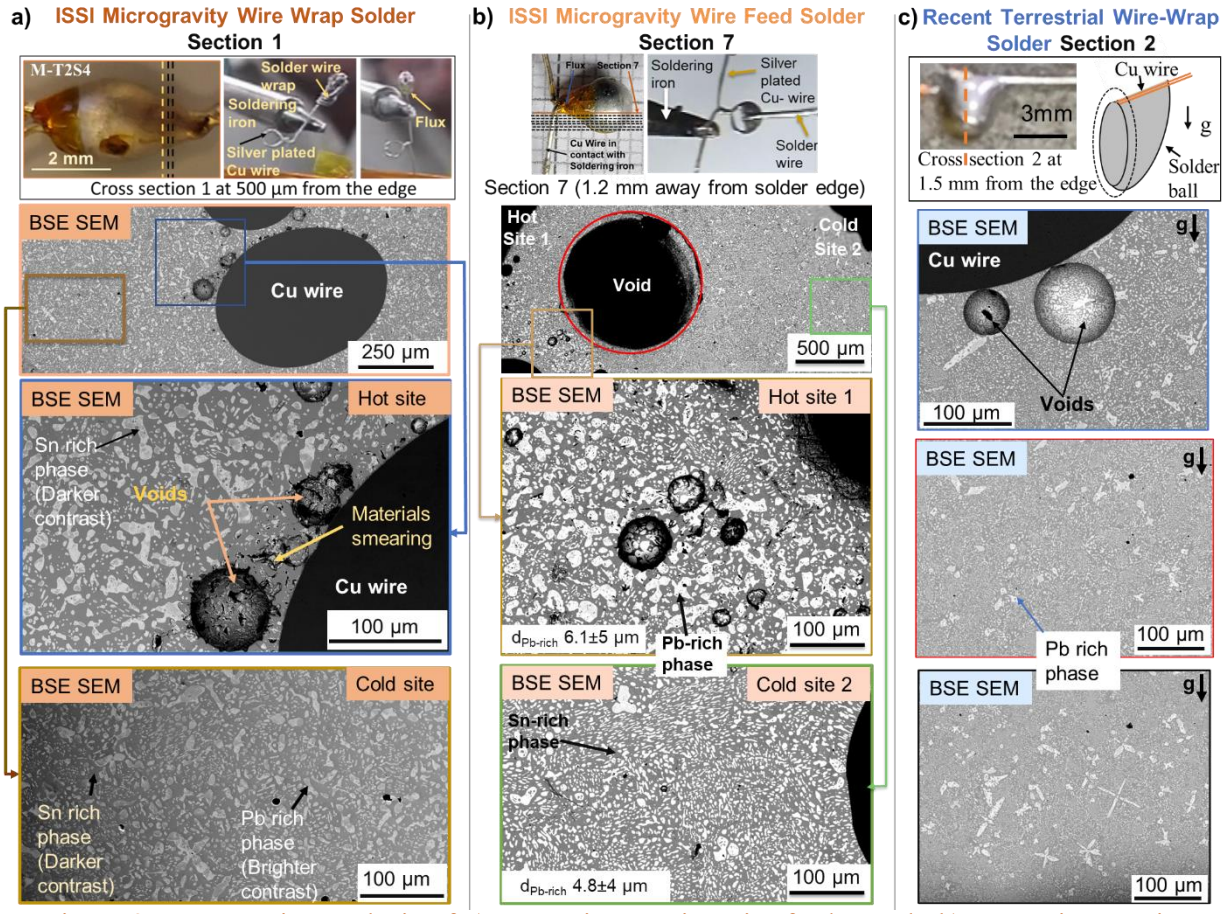


Figure 3: Comparative analysis of a) ISSI microgravity wire-feed sample b) ISSI microgravity wire wrap sample c) Recent terrestrial wire-wrap solder. Microgravity wire feed solder demonstrate increased void and Pb-rich dendrites accumulation than Microgravity wire wrap and terrestrial wire wrap.

The effect of this Marangoni convection is less significant in terrestrial conditions due to dominating natural convection. However, the effect of Marangoni flow becomes more significant in microgravity conditions where the bulk natural convection is eliminated. Therefore, to further

characterize the effects of the lack of Earth's natural convective flow and buoyancy effects on the microgravity solder microstructure, we performed destructive serial sectioning (Fig. 3a-c). Microstructural characterization demonstrates the presence of the voids close to the Cu wire in ISSI microgravity 40Pb-60Sn solders in wire wrap and wire feed configurations (Fig. 3a), which may have been at a higher temperature due to the direct contact with the soldering iron.

This higher number of voids near the Cu wire was driven by the presence of the thermocapillary flow. In addition, The ISSI microgravity 40Pb-60Sn wire wrap (Fig. 1) solder demonstrated a homogeneous distribution of Pb-rich phase (Fig. 3a). In contrast to the ISSI microgravity 40Pb-60Sn wire wrap solder, ISSI microgravity 40Pb-60Sn wire feed solder (Fig. 1) demonstrated the accumulation of voids and Pb-rich dendrites in one region (site 1 in Fig. 3b). The average Pb-particle size near site 1 was 6.1 ± 5 µm and near site 2 (expected to be at a higher temperature) was 4.8 ± 5 µm. This difference in the solder microstructure is probably caused by the strong Marangoni flow in the wire feed solder. Further characterization of this sample revealed that this Marangoni effect was probably caused by the presence of a large void (d_{void} = 1.1 mm) in the ISSI microgravity wire feed solder (further discussion in the next section). Additionally, the eutectic lamellae spacing near the hot site was 2.15 ± 0.45 µm, and near the cold site was 1.89 ± 0.43 µm. A slower cooling rate probably caused this lower eutectic lamella spacing at the hot site than at the cold site. In comparison, recently prepared terrestrial solder demonstrated the inhomogeneous distribution of Pb-rich dendrites caused by Earth's gravity, where Pb-rich dendrites (heavier phase) accumulated at the bottom of the solder ball (Fig. 3c).

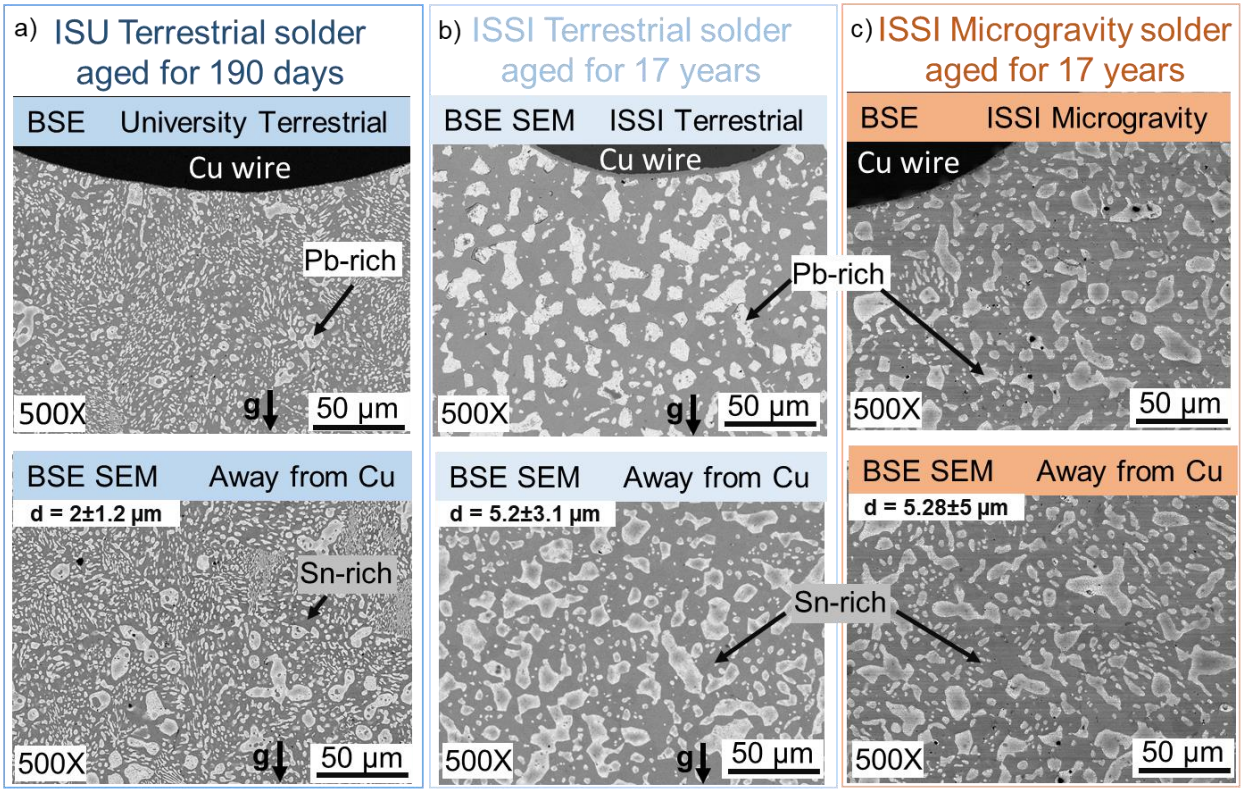


Figure 4: Microstructure comparison of a) ISU terrestrial b) ISSI terrestrial c) ISSI microgravity 40Pb-60Sn solder. ISSI solders demonstrate a significant amount of coarsening caused by long-term aging than ISU 40Pb-60Sn solder.

Comparison between the microstructure of ISU terrestrial, ISSI microgravity, and ISSI terrestrial 40Pb-60Sn solder (Fig. 4) demonstrates the significant effect of aging on the Pb-rich particle size in ISSI solders. The coarser eutectic structure observed in both ISSI microgravity and ISSI terrestrial solder is attributed to a similar aging time period.

Outcomes from Year 2

In the **Year 1 report**, we highlighted how the Lack of buoyancy force and natural convection causes increased porosity/voids in the ISSI microgravity 40Pb-60Sn solder and how void-induced Marangoni flow impacts the microstructure of solder in microgravity conditions. In the **Year 2 final report**, we focus on demonstrating how the resultant microstructure, influenced by void-induced Marangoni flow, affects the mechanical properties of the ISSI microgravity 40Pb-60Sn solder. This analysis underscores the critical importance of understanding and mitigating the effects of void-induced Marangoni flow on solder reliability in microgravity environments.

Task D: Micro-Mechanical Testing Strategies for Solders at Elevated and Cryo Temperatures

Effect of microstructure, influenced by void-induced Marangoni convection, on microgravity solder properties:

As discussed in Task C, void-induced Marangoni convection in ISSI microgravity 40Pb-60Sn wire-feed solder drives the accumulation of voids and Pb-rich dendrites toward the hot site. This microstructural variation significantly impacts the mechanical properties of the solder. We performed Berkovich nanoindentation experiments at 1.5 µm indentation depth using the Hysitron TI950 nanoindenter. Nanoindentation analysis of the wire feed solder reveals a reduction in hardness from 0.15 GPa at the cold site to 0.11 GPa at the hot site (Fig. 5), correlating with the localized accumulation of the softer Pb-rich phase near the substrate. This uneven distribution of Pb-rich dendrites and resultant lower hardness at the solder joint, driven by void-induced Marangoni convection, poses a risk to the structural integrity of solder joints during in-space soldering of electronic components. To ensure reliable performance and achieve a homogeneous microstructure, it is crucial to mitigate the effects of void-induced Marangoni convection on void and dendrite accumulation. This understanding underscores the importance of designing soldering processes that minimize these microstructural inconsistencies in microgravity environments.

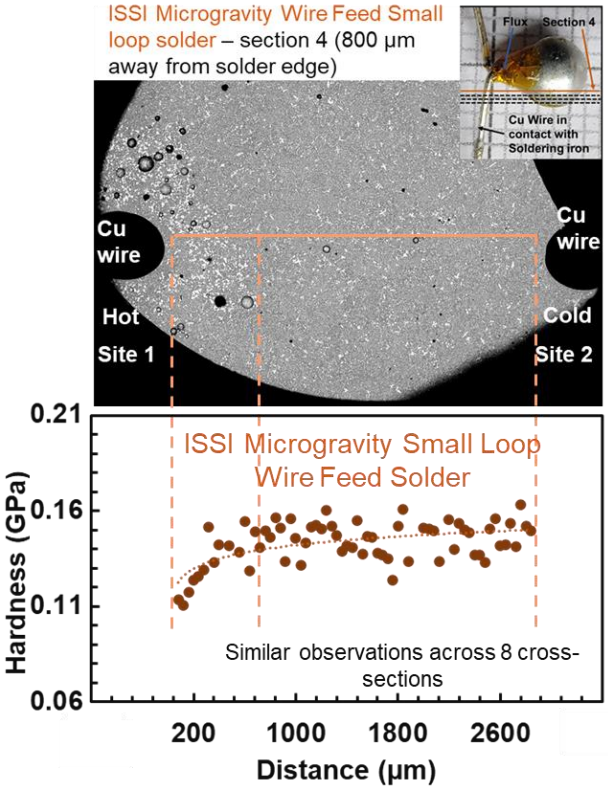


Figure 5: Influence of Pb-rich dendrite accumulation on solder hardness from the cold site to the hot site. The hardness decreases toward the hot side due to the accumulation of softer Pb-rich dendrites caused by void-induced Marangoni convection.

Characterizing the Effects of the Extreme Elevated and Cryogenic Temperature on Solder-joint Properties:

Long-duration space exploration missions such as Mars rover missions have traditionally kept their electronics in a heated enclosure known as a "warm electronics box" (WEB) to protect them from the extreme cryo temperature on Mars [3]. However, future Mars missions are planning to use a distributed, non-heated architecture for the rover's electronics, where the components will be placed near the systems they interact with rather than in a heated enclosure [3]. The new approach relies on low-cost, commercially available, and space-qualified electronic packaging assemblies. However, these components must be capable of operating in a functional range of – 120ºC to +25ºC for the Mars missions, – 230ºC to +120ºC for the missions to Moon, and – 157ºC to +121ºC outside ISS. Therefore, this project focuses on studying the properties of 1) Pure Tin, 2) 40Pb-60Sn, 3) 50Pb-50Sn, and 4) 95.5Sn-3.8Ag-0.7Cu solder alloys at a) RT, b) Elevated temperatures, and c) Cryo temperatures.

For detailed discussion on aging properties of terrestrial 40Pb-60Sn solder, please refer to the year 1 report (Link).

Investigate the effect of aging on microstructure and properties of ISU terrestrial 50Pb-50Sn solders under ambient conditions:

Sample analyzed: ISU Terrestrial 50Pb-50Sn Samples 1, 2, and 3

Future off-eutectic 50Pb-50Sn samples produced aboard the ISS will take many months to return to the Earth for testing. Due to the high homologous temperature ($T_{RT}/T_M = 0.60$) for 50Pb-50Sn

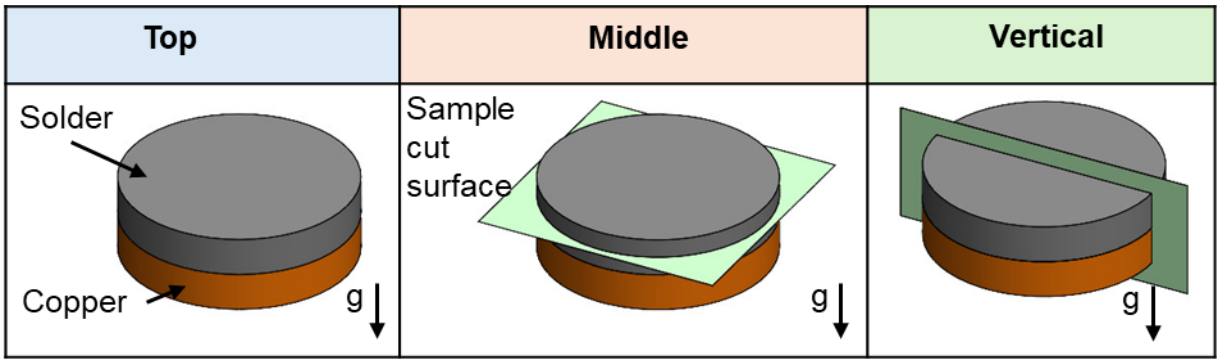


Figure 6: Illustration of cut planes for the three samples used in off-eutectic aging. Top, middle, and vertical sections were analyzed to assess the effect of gravity on the heavier Pb-rich phase and its impact on aging properties.

solder (T_M = 183ºC), classical elemental diffusion described by Fick's law initiates the grain coarsening [17] that can affect the performance and reliability of these solder joints for space applications. Since aging happens in the solid state of the samples, aging behavior in microgravity solder is expected to be the same as terrestrial solder. Therefore, to account for the effect of aging in microgravity, we prepared fresh, terrestrial off-eutectic solder samples to study the rate and rate and effect of aging on this composition.

Three samples were prepared by melting small solder disks into copper coupons in a furnace. Once removed, the samples were allowed to cool to room temperature and were polished to three different faces: top, middle, and vertical (Fig 6). Three different faces were chosen to determine the effects of gravity on aging. Lead is denser than tin, so we expected to see an increase in lead concentration near the bottom of the sample, which likely leads to slightly different aging. Additionally, the effects of the copper substrate were measured by dividing the top and middle samples into two regions: the center and the edge.

Testing was done using nanomechanical indentation tests over the course of 494 days in addition to SEM imaging to record the grain size evolution. Every testing day involved performing 36 indentation tests in each region to measure local surface hardness. Additionally, the samples were imaged using an SEM to record where the previous day's indents landed as well as take images for grain size measurements. In each sample region, at least three images were captured for grain size analysis. Additionally, images were taken at control points to visually track grain coarsening within the same region throughout the aging study. Testing was conducted daily for the first 15 days after solidification, every three days from days 15 to 60, and every six days from days 60 to 90.

Nanomechanical testing revealed a hardness reduction in the top sample from 0.18 GPa on day 1 to 0.15 GPa by day 60, while the middle section decreased from 0.17 GPa to 0.14 GPa over the same period (Fig 7). Additionally, the hardness decreased the fastest over the first 15 days after solidification followed by a gradual decrease in the hardness.

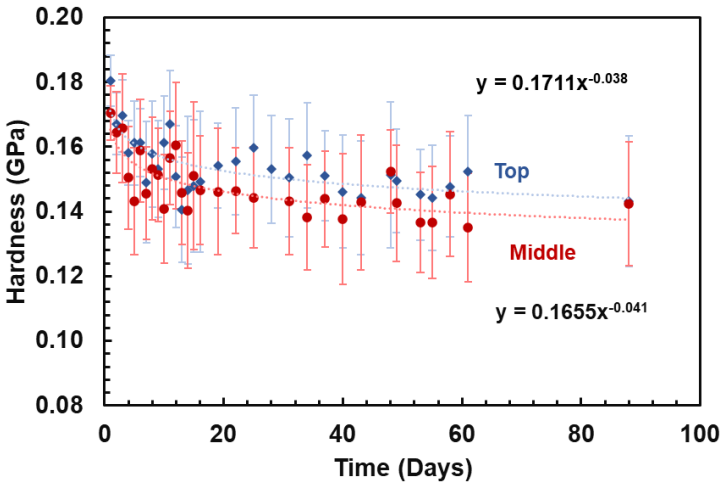


Figure 7: Impact of aging on the hardness of terrestrial off-eutectic 50Pb-50Sn solder. Hardness decreases significantly due to room-temperature aging, with a rapid decline in the first 15 days followed by a gradual reduction. The middle section exhibits lower average hardness than the top section due to the accumulation of the softer Pb-rich phase caused by gravity.

Grain coarsening was observed in both eutectic and pro-eutectic regions, affecting both Pb-rich and Sn-rich phases (Fig. 8). In addition to grain growth, Sn-rich precipitates formed within Pb-rich regions due to the significantly lower solubility of Sn in Pb at room temperature compared to the eutectic temperature.

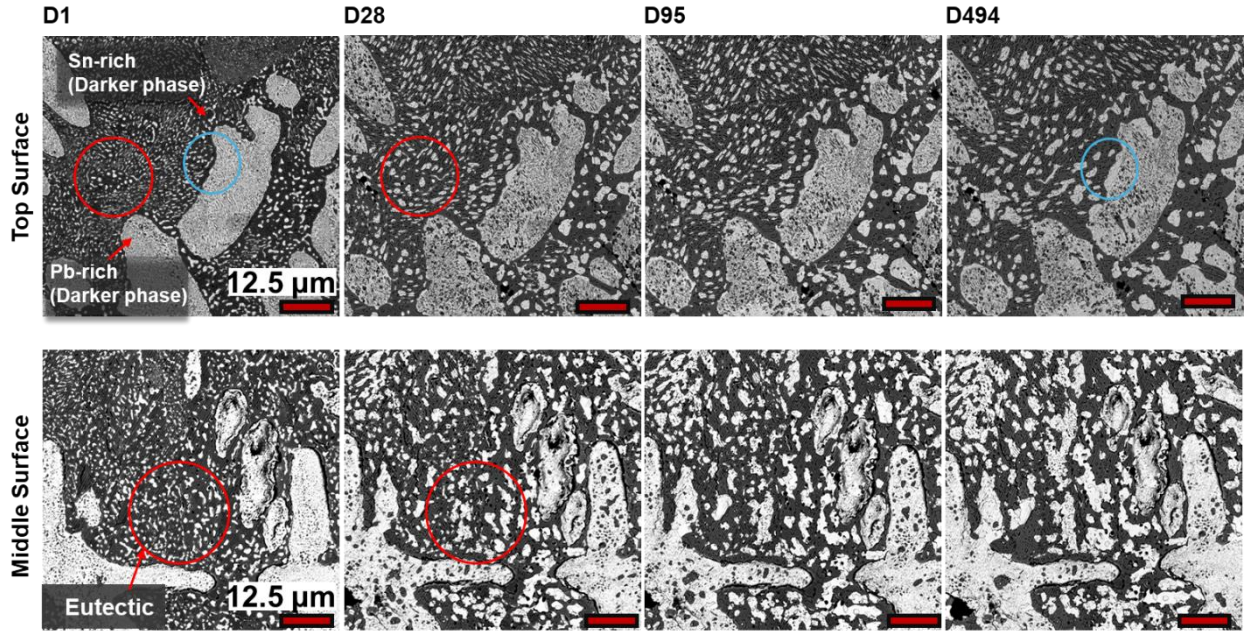


Figure 8: Impact of aging on the microstructure of the top and middle sections. Significant grain growth is observed in the eutectic Pb-rich phase, along with Sn-rich precipitation within the Pb-rich pro-eutectic dendrites. Red circles highlight grain growth within the eutectic region, while blue circles indicate grain growth along the edge of a Pb-rich dendrite.

The rate of grain coarsening was quantified by measuring the average area of Pb-rich phases within the eutectic structure and calculating the equivalent circular diameter (Fig. 9). Figure 9 illustrates the effect of aging on particle size within the eutectic region of terrestrial 50Pb-50Sn solder. In the top section, particle size increased from 0.46 µm on day 1 to 0.73 µm on day 87, while in the middle section, it grew from 0.45 µm to 0.90 µm over the same period. Similar to the hardness response, Pb-rich particle size exhibited rapid growth during the first 15 days, followed by a more gradual increase. Additionally, the higher standard deviation results from the wide distribution of particle sizes, as shown in Figure 10. The distribution plots indicate an increased frequency of larger particles and a decreased frequency of smaller ones, as indicated by two distinct peaks in

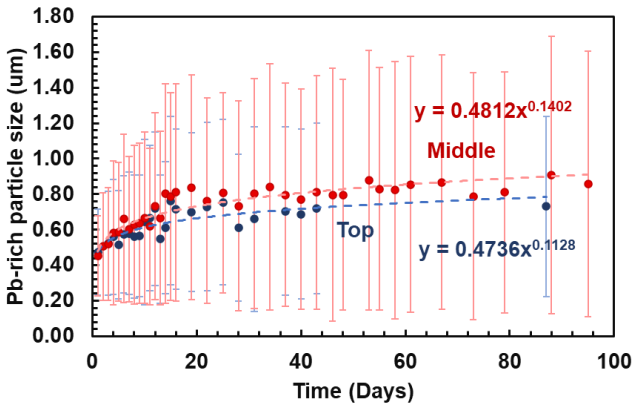


Figure 9: Impact of aging on Pb-rich particle size. Pb-rich particles grow over time, with the middle section exhibiting a larger average particle size than the top section. This difference is attributed to the gravitational accumulation of the Pb-rich phase in the middle section.

the normal distribution. This trend suggests that larger particles grow at the expense of smaller ones.

Additionlally, Pb-particle size follows a power-law relationship in both the top and middle

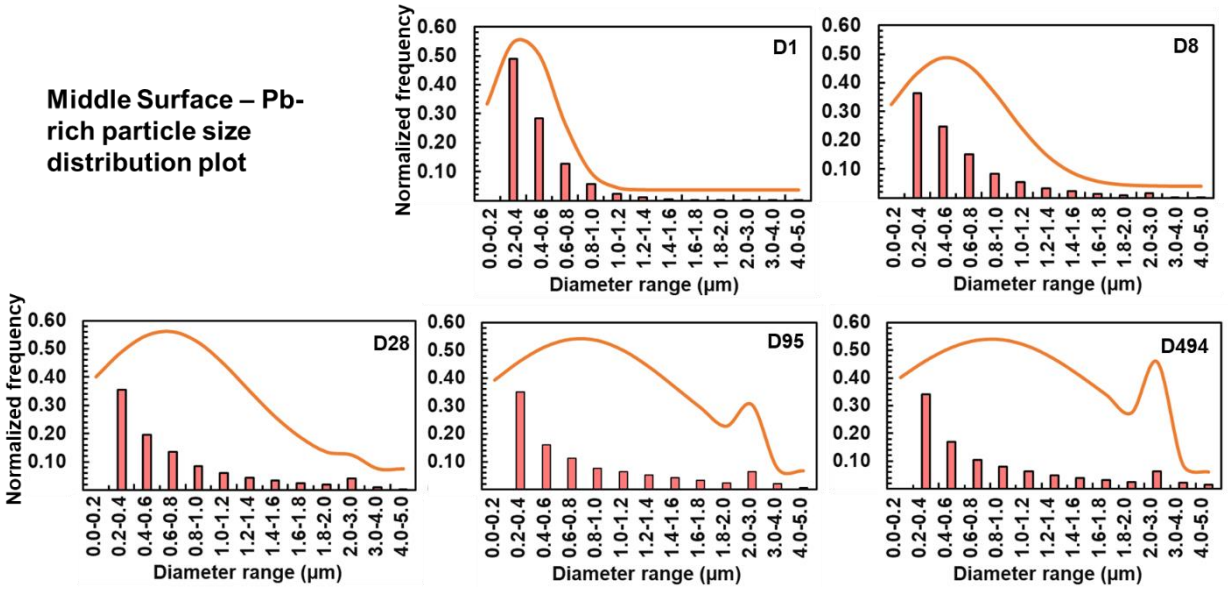


Figure 10: Effect of aging on Pb-rich particle size distribution over time in the middle section. The distribution plot reveals an increase in larger particles and a decrease in smaller ones, as indicated by two peaks in the normal distribution. This trend suggests that larger particles grow at the expense of smaller ones.

sections, with the middle section exhibiting a larger average Pb-rich particle size. The constant K for the top and middle sections is 0.4736 μm and 0.4812 μm, respectively, while the power exponent n is 0.1128 for the top section and 0.1402 for the middle section. The larger average Pb-rich particle size in the middle section is attributed to the accumulation of the heavier Pb-rich phase caused by pull of gravity, leading to a higher Pb concentration compared to the top section. Furthermore, Pb particle size within the eutectic region follows a distinct scaling behavior similar to the Hall-Petch relationship (Fig. 11), with the H_o values of 0.095 GPa for top section and 0.073 GPa for the middle section. The scaling constants are K_{Top} = 0.047 GPa($\mu m^{0.5}$) and K_{Middle} = 0.062 GPa($\mu m^{0.5}$). The middle section exhibits a slightly different scaling behavior compared to the top section, likely due to a higher concentration of the Pb-rich phase.

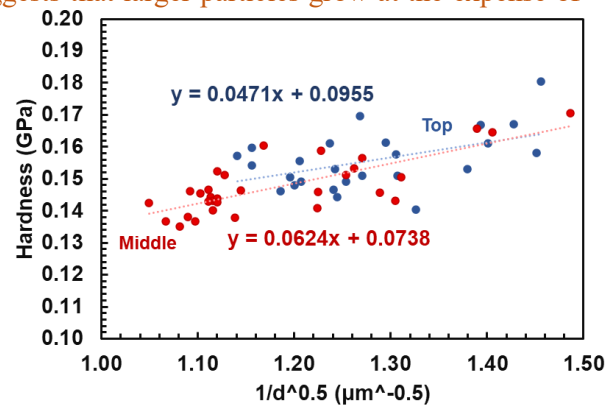


Figure 11: Relationship between Pb-rich particle size and hardness in terrestrial 50Pb-50Sn solder. The middle section exhibits a slightly different scaling behavior compared to the top section, likely due to a higher concentration of the Pb-rich phase.

Investigating the Effect of the cryogenic temperature on β-Sn to α-Sn phase transformation, ductile to brittle transition and resultant properties of the solder joints: Sn-based solders undergo a ductile to brittle transition (DBT) at the cryo temperature leading to brittle crack formation, which can cause the catastrophic failure of the solder joints. Additionally, Sn-based solders generally undergo a β-Sn to α-Sn phase transition below 13ºC, where β-structure (Body-centered tetragonal) transforms to α-structure (Diamond cubic). To investigate the effect of various cryo temperatures (similar to temp conditions on the Moon, Mars, and outside ISS) on solder properties, we performed high-throughput spherical indentation on terrestrial 40Pb-60Sn and pure Sn(111) single crystals at RT and -120ºC and in-situ SEM micropillar compression on terrestrial 40Pb-60Sn, 50Pb-50Sn, and 95.5Sn-3.8Ag-0.7Cu solder compositions at RT, – 40ºC, – 85ºC, – 120ºC and – 145ºC. These temperatures are selected based on the literature review [5, 6, 10] on Sn phase transformation and solder joint application during space exploration missions.

Solder properties under extreme temperatures using Spherical indentation:
Sample analyzed: Sn(111) single crystal and ISU terrestrial 40Pb-60Sn solder

To investigate the long-term reliability of the solder joints under various elevated and cryo temperature conditions, we utilized the Hysitron TI 980 nanoindenter to obtain the nanoindentation stress-strain from Sn single crystal and terrestrial 40Pb-60Sn solder at RT and - 120ºC. Our novel high-throughput spherical nanoindentation stress-strain data-analysis protocols [30-32] which allow us to measure both the elastic (e.g. modulus) and plastic (indentation yield Y_{ind} (Fig. 10) and strain hardening coefficients) properties from indentation zones ranging in size from ~30 nm to 10 μm [30, 31]. By using an indentation zone size smaller than the critical feature size [33], we can measure changes in Y_{ind} as a function of orientation [31, 34]. Similarly, by using a sufficiently large-sized indenter, we propose to interrogate the bulk properties (e.g. combined response of Pb-rich and Sn-rich phases) of the solder as a function of temperature.

The spherical nanoindentation stress-strain protocols [30-32] shown in Fig. 12 offer certain distinct advantages, such as the non-destructive nature of indentation experiments, ease of experimentation (only a flat surface prior to indentation is needed), high throughput, and versatility. More importantly, these protocols allow the analysis of the initial loading segments of spherical indentation – before the indentation itself imposes additional local plastic deformation and alters the local material state and its properties. These indentation data analysis methods have been used successfully to capture the local loading and unloading elastic moduli, the local indentation yield strengths, and certain aspects of post-yield strain hardening behavior in various polycrystalline metal samples, and across their grain boundaries.

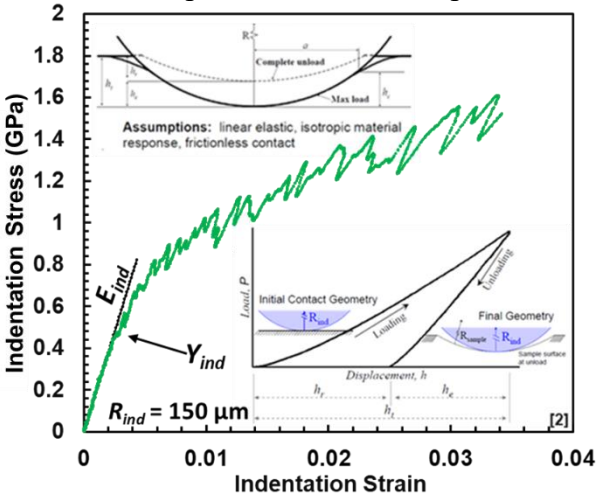


Figure 12: Spherical Indentation Stress-Strain curve and a typical measured spherical indentation load-displacement curve with the initial and final contact geometries.

Extraction of indentation stress-strain curves from the measured load-displacement data (described in Ref. 30) can be briefly

summarized as a two-step procedure. The **first step** in the data analysis process is determining of the point of effective initial contact in the given data set, i.e. a clear identification of a zero-point that makes the measurements in the initial elastic loading segment consistent with the predictions of Hertz's theory[35-37]. As shown in Ref. 30, the zero point can be conveniently determined using the following equation for the initial elastic segment in a frictionless, spherical indentation:

$$S = \frac{3P}{2h_e} = \frac{3(\tilde{P}-P^*)}{2(\tilde{h}_e-h^*)} \qquad (1)$$

where $\tilde{P}$, $\tilde{h}_e$ and S are the measured load signal, the measured displacement signal, and the continuous stiffness measurement (CSM) signal in the initial elastic loading segment from the machine, respectively, and P^* and h^* denote the values of the load and displacement values at the point of effective initial contact. Rearrangement of Eq. (1) reveals that a plot of $\left(\tilde{P} - \frac{2}{3}S\tilde{h}_e\right)$ against S will produce a linear relationship whose slope is equal to $\left(-\frac{2}{3}h*\right)$ and the y-intercept is equal to P^* . A linear regression analysis can then be performed to identify the point of the effective initial contact (P^* and h^*) very accurately.

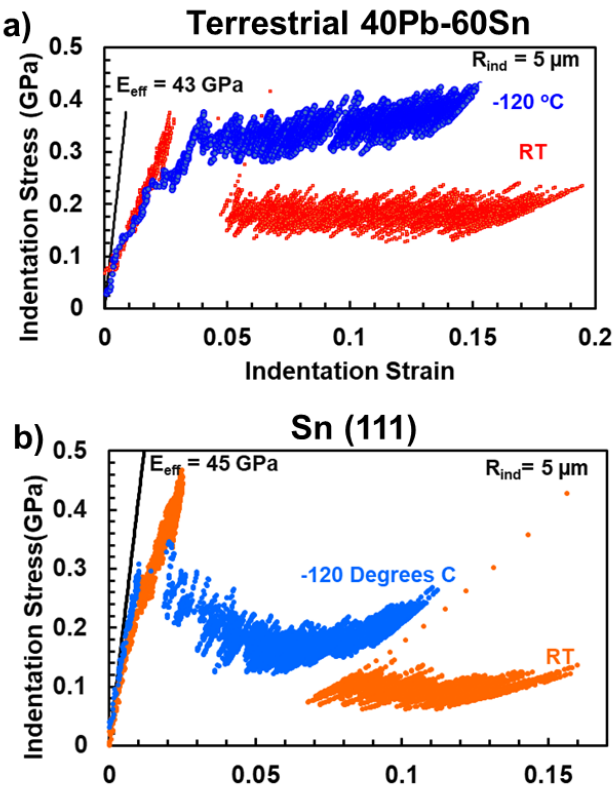


Figure 13: Spherical indentation stress-strain curves for (a) terrestrial 40Pb-60Sn solder and (b) pure Sn (111) single crystal. The 40Pb-60Sn solder exhibits increased strength at cryogenic temperatures, while the Sn single crystal shows softening, likely due to the β-Sn to α-Sn phase transformation at low temperatures in Sn single crystal.

In the **second step**, the values of indentation stress and strain can be calculated by recasting Hertz theory for frictionless, elastic, spherical indentation as

$$\sigma_{ind} = E_{eff}\varepsilon_{ind}, \quad \sigma_{ind} = \frac{P}{\pi a^2}, \quad \varepsilon_{ind} = \frac{4}{3\pi}\frac{h_t}{a}$$

$$a = \frac{S}{2E_{eff}}, \quad \frac{1}{E_{eff}} = \left(\frac{1-\nu^2}{E}\right)_{sample} + \left(\frac{1-\nu^2}{E}\right)_{indenter}, \quad \frac{1}{R_{eff}} = \frac{1}{R_i} + \frac{1}{R_s}, \qquad (2)$$

where σ_{ind} and ε_{ind} are the indentation stress and indentation strain, a is the radius of the contact boundary at the indentation load P, h_e is the elastic indentation depth, S (= dP/dh_e) is the elastic stiffness described earlier, R_{eff} and E_{eff} are the effective radius and the effective stiffness of the indenter and the specimen system, ν and E are the Poisson's ratio and Young's modulus, and the subscripts s and i refer to the specimen and the indenter, respectively.

We performed indentation experiments using R_{ind} = 5-micron spherical probe. The load and displacement signal was converted into Indentation-Stress Strain using the protocol mentioned earlier.

For the terrestrial 40-60 lead-tin solder, an increase in strength is observed at cryogenic temperatures where 0.05 offset yield strength increased from 0.18 GPa at RT to 0.32 GPa at -120°C (Fig 13a). However, pure tin exhibits significant softening at -120°C (Fig. 13b), which is an unusual behavior in indentation experiments. While this phenomenon is hypothesized to be related to the phase transformation of tin, further detailed analysis is required to confirm this assumption. Notably, such softening behavior is not observed in lead-tin solder alloys. To further characterize the effect of cryogenic temperature on solder properties, we performed in-Situ SEM micro-pillar compression on ISU terrestrial 40Pb-60Sn and 95.5Sn-3.8Ag-0.7Cu solders, as discussed in the next section.

Micro-pillar fabrication and In-situ SEM compression experiments: We fabricated the micropillars (Fig. 14a) on terrestrial 40Pb-60Sn and 95.5Sn-3.8Ag-0.7Cu solders using the micro-EDM at the

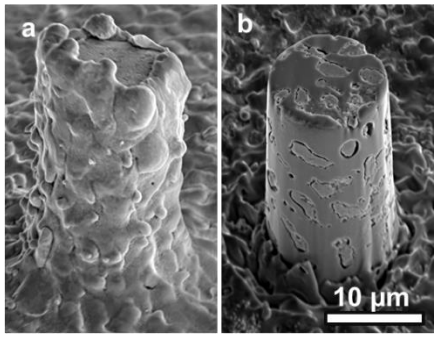


Figure 14: a) rough-cut 40Pb-60Sn micropillar using the micro-EDM b) final pillar after the pillar cleaning using the focused ion beam milling at very low current (0.85 nA).

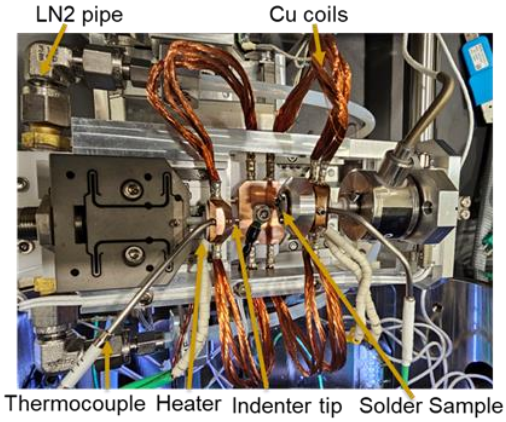


Figure 15: Alemnis in-Situ SEM nanoindenter used for micro-pillar compression experiments at cryogenic temperatures.

SmalTec international (Lisle, IL 60532). The rough micropillars were cleaned (Fig. 14b) using the focused ion beam milling at a very low current (0.85 nA) in order to reduce the deformation zone from the micro-EDM process. The micropillars were then compressed at the loading rate of 75nm per second at room and various cryogenic temperatures using the Alemnis in-situ SEM cryo indenter (Fig. 15). During the compression, load, displacement and time were recorded with simultaneous SEM imaging and video recording. The load-displacement was converted into the stress-strain, and the 0.2% offset yield strength was calculated.

Ductile to Brittle Transition in Terrestrial 40Pb-60Sn at cryo temperatures:

Sample analyzed: ISU terrestrial 40Pb-60Sn solder sample 10

The load-displacement data were converted into engineering stress-strain curves (Fig. 16 a) to calculate the plastic strain at instability, 0.2% offset yield strength, and stress at instability for each testing temperature. Notably, there is a significant decrease in the plastic strain at instability (Fig. 16 b) below -85°C. Two potential factors could explain this behavior:

1. First, a ductile-to-brittle transition and the β-to-α phase transformation in Sn. Based on current observations, the primary cause is likely the ductile-to-brittle transition. This conclusion is supported by several findings: first, an increase in stress at instability (Fig. 16c) is observed

from 45.9 MPa at RT to 106.5 MPa at -40°C, 129.1 MPa at -85°C, 129.1 MPa at -120°C, and 161.4 MPa at -145°C, which contradicts the expected decrease in strength associated with a β-Sn to α-Sn phase transformation in Sn-based solders. The plastic strain at instability decreased from 0.109 at RT to 0.062 at -85°C and reached a saturation level below -85°C, further demonstrating the effect of ductile to brittle transition.

2. Second, preliminary XRD analyses do not indicate any evidence of the formation of the α-Sn phase caused by a phase transformation.
3. Third, continuous imaging during experiments shows that deformation at cryogenic temperatures is concentrated along the boundary between Pb-rich and Sn-rich phases (Fig. 17), a behavior consistent with ductile-to-brittle transitions. However, a definitive confirmation of this conclusion requires further investigation using TEM analysis.

Ductile to Brittle Transition in Terrestrial 95.5Sn-3.8Ag-0.7Cu Solder at cryo temperatures:

Sample analyzed: ISU terrestrial 95.5Sn-3.8Ag-0.7Cu solder sample 8

Similar to ISU terrestrial 40Pb-60Sn solder, the load-displacement data from micropillar compression experiments (Fig. 18) on terrestrial 95.5Sn-3.8Ag-0.7Cu solder were converted into engineering stress-strain curves to calculate the plastic strain at instability, 0.2% offset yield strength, and stress at instability for each testing temperature. The microstructural image in Figure 15 highlights distinct phase distributions, including Sn-rich dendrite, Ag_3Sn + Sn eutectic, and Cu_6Sn_5 intermetallic phases, which may play a role in the observed mechanical behavior at different temperatures.

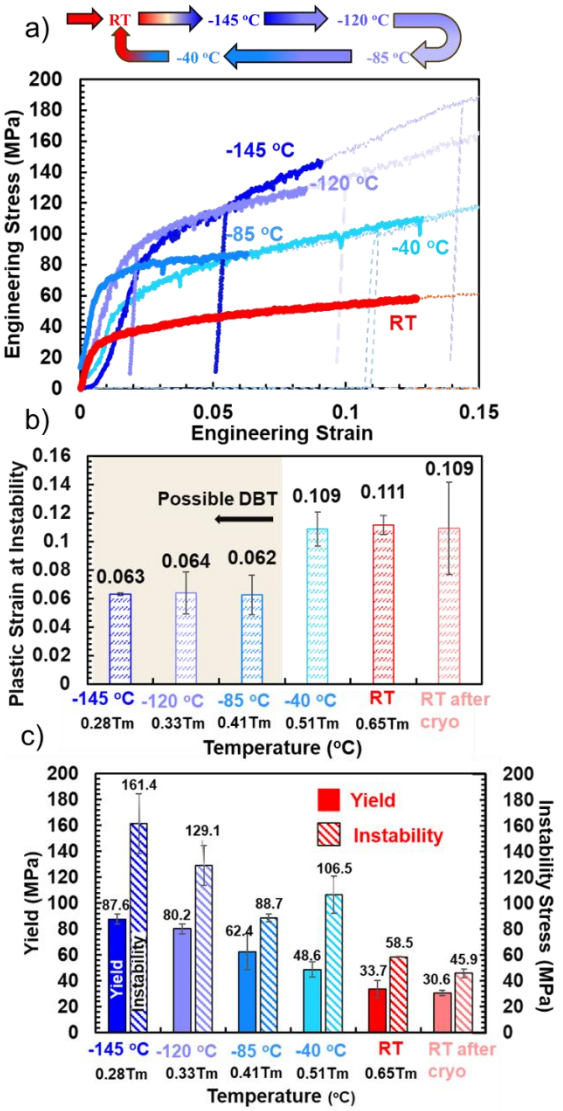


Figure 16: (a) Engineering stress-strain curves for 40Pb-60Sn solder tested at -145°C, -120°C, -85°C, -40°C, and room temperature (RT). (b) Plastic strain at instability as a function of temperature. (c) 0.2% offset yield strength and stress at instability. Drop in plastic strain from $\varepsilon = 0.109$ at RT to $\varepsilon = 0.06$ below -85°C, suggest a ductile-to-brittle transition near this temperature.

Figure 19 presents the average response of micropillars, including average yield strength, average stress at instability, and average plastic strain at instability, irrespective of the pillars' location (Fig. 19a) on the sample or the phase fraction within each pillar. Figure 18 illustrates the microstructural morphology distribution of solder joints after reflow. Long-strip and granular Ag_3Sn, along with massive Cu_6Sn_5 intermetallic compounds (IMCs), are randomly and sparsely distributed within the β-Sn matrix. The results

presented in Fig. 19 reveal several key insights into the mechanical behavior of solder under varying temperature conditions.

- The ISU terrestrial 95.5Sn-3.8Ag-0.7Cu solder exhibited significantly higher strength (Fig. 19b & c) at cryogenic temperatures compared to room temperature, as observed from the stress-strain curves. The stress at instability increased from 78.1 MPa at RT to 113.4 MPa at -

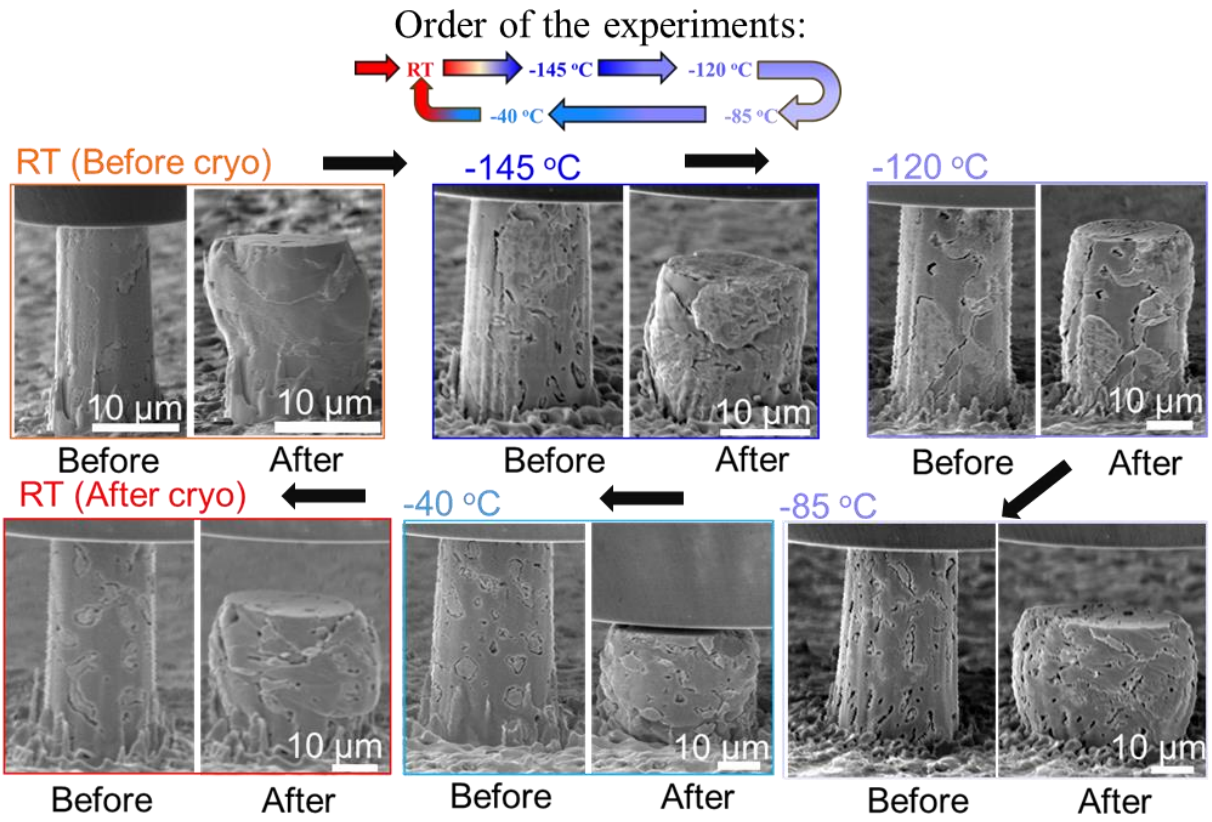


Figure 17: SEM images of micropillars before and after the 40Pb-60Sn micropillar compression. Deformation becomes concentrated along the boundary between Pb-rich and Sn-rich phases below -85ºC probably caused by ductile to brittle transition.

40°C, 94.2 MPa at -85°C, 153.8 MPa at -120°C, and 153 MPa at -145°C. Similarly, the 0.2% offset yield strength increased from 49.5 MPa at RT to 78.9 MPa at -40°C, 78.3 MPa at -85°C, 126.1 MPa at -120°C, and 103 MPa at -145°C.

- There is noticeable saturation in the average instability stress at around 78 MPa below -85°C, indicating a potential limit to the strengthening effect at lower temperatures caused by ductile to brittle transitions.
- The average plastic strain at instability dropped from 0.06 at RT to 0.04 at -40°C, 0.03 at -85°C, 0.03 at -120°C and 0.02 at -145°C. Notably, the plastic strain at instability shows saturation at 0.03 below -85°C, with an overall reduction in the plastic strain at these cryo temperatures (Fig. 19d). However, the plastic strain at instability for each pillar group demonstrated a different trend than the average values as discussed in this section.
- The larger standard deviation in plastic strain at instability at RT is caused by the difference in amount of different phases in each pillar as discussed later in this section.

Two potential factors could explain these insights from micropillar compression on ISU terrestrial 95.5Sn-3.8Ag-0.7Cu solder under cryogenic temperatures: (1) a ductile-to-brittle transition and (2) the β-to-α phase transformation in tin. Based on current observations, the primary cause is likely the ductile-to-brittle transition.

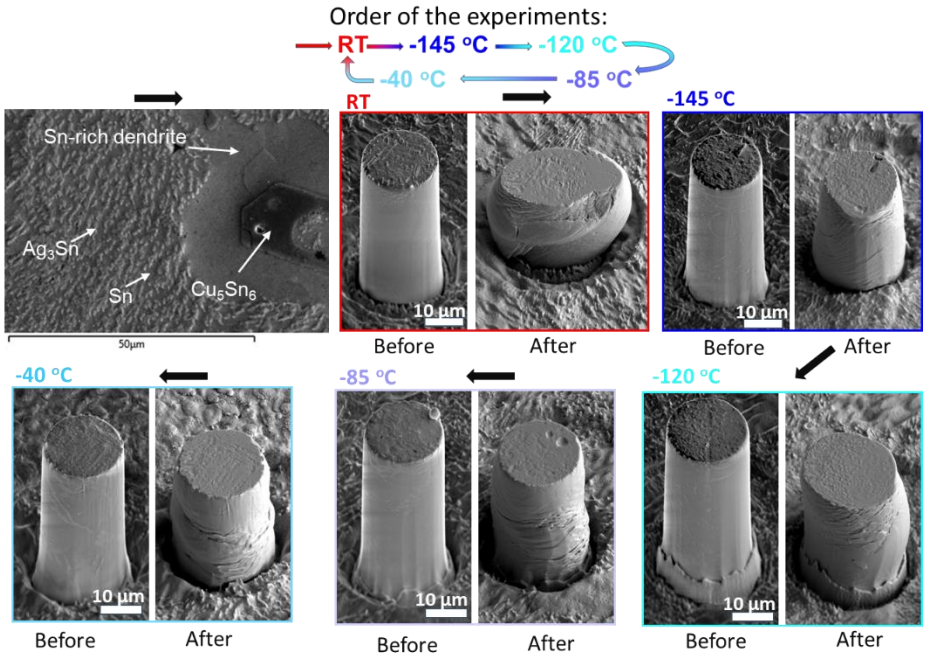


Figure 18: Micropillar compression on 95.5Sn-3.8Ag-0.7Cu solder. Deformation becomes concentrated along the different phase boundaries depending on the temperature and phase fraction signaling the ductile to brittle transition.

This conclusion is supported by the following finding: an

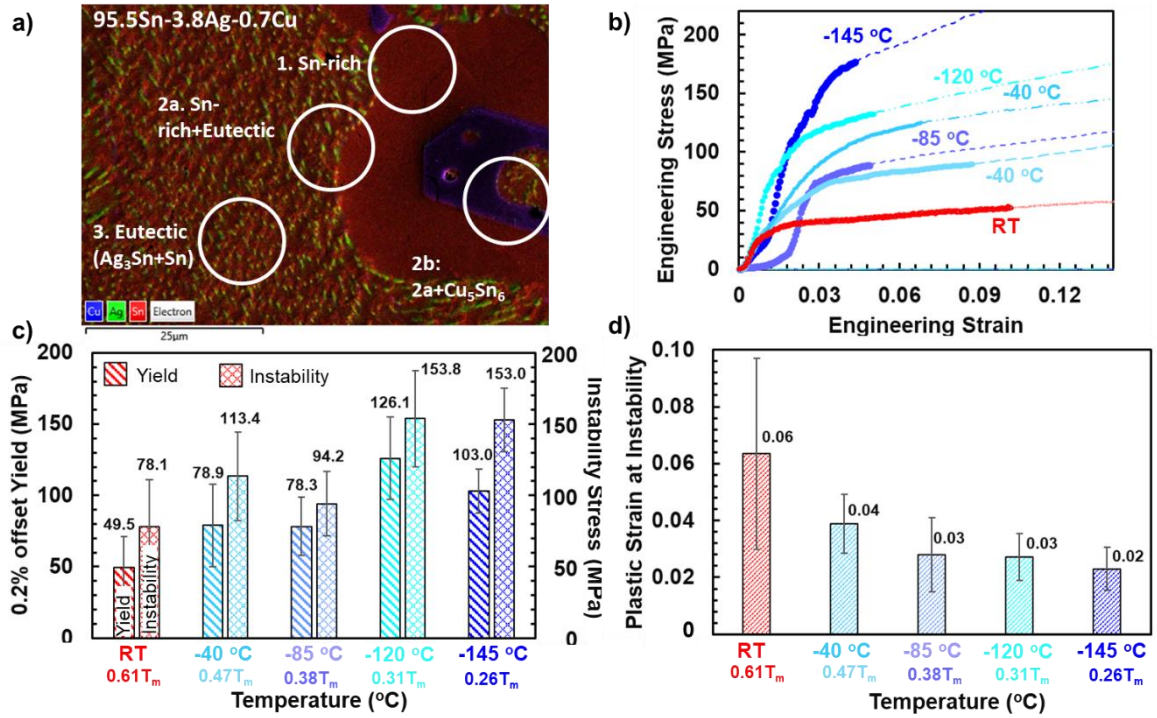


Figure 19. a) Three distinct categories of pillar groups, b) Engineering Stress-Strain curve for 95.5Sn-3.8Ag-0.7Cu tested at -145 °C, -120 °C, -85 °C, -40 °C, and RT, c) Yield strength and stress at instability, d) Plastic Strain at instability. Reduction at saturation in the plastic strain at instability is probably caused by ductile to brittle transition.

increase in strength is observed at cryogenic temperatures, which contradicts the expected decrease in strength associated with a phase transformation. Furthermore, the saturation in plastic strain at temperature at and below -85°C is attributed to a ductile-to-brittle transition, as evidenced by the concentration of deformation along specific phase boundaries in the microstructure.

The data were further categorized based on the location of the micropillars on the sample, grouping them into three distinct categories for each testing temperature, with Group 2 further divided into two subsets (Fig. 20). The groups are defined as follows: Group 1) Predominantly Sn-rich – pillars containing approximately 90% Sn-rich phase which exists only for -40 °C, Group 2a) A mixture of Sn-rich phase and Ag_3Sn + Sn eutectic, Group 2b) A mixture of Sn-rich phase, Ag_3Sn + Sn eutectic, and Cu_6Sn_5 intermetallic phases, and Group 3) Predominantly Ag_3Sn + Sn eutectic.

Figure 20 illustrates the various deformation mechanisms observed in different groups, which vary based on the testing temperature. In Group 1 at -40 °C, deformation occurs primarily through the Sn-rich dendrites (Table 4). In Group 2a, during micro-pillar compression experiments, deformation at -85°C and higher temperatures progresses along the boundary between the Sn-rich dendrites and

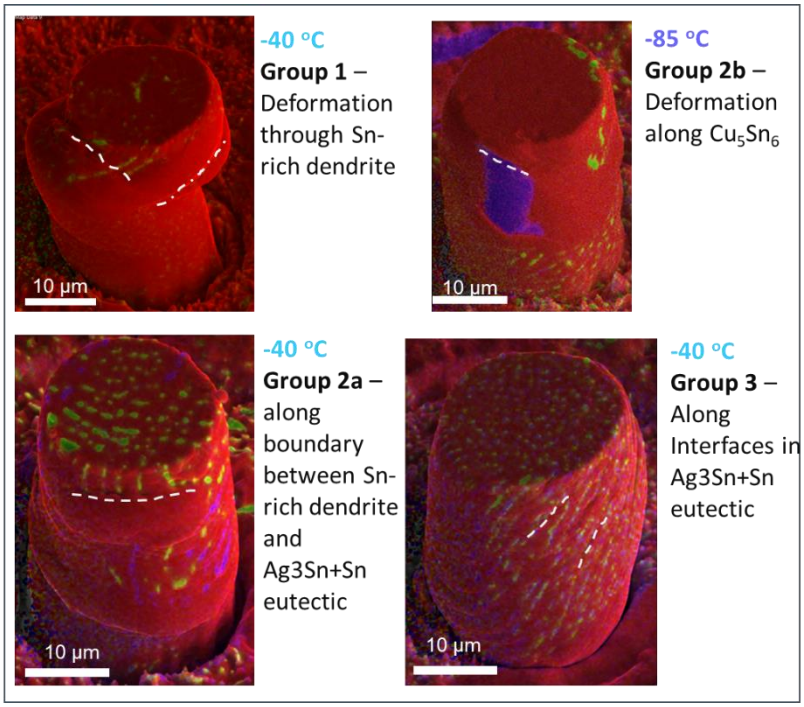


Figure 20. Deformation behavior observed in different groups at -40 °C and -85 °C. The deformation behavior changes depending on the assigned group and testing temperature.

Table 4. Deformation initiation point during micropillar compression at cryo temperatures in various groups. Deformation behavior varies based on the testing temperature.

Group	1	2a	2b	3
			DBT	
RT ($0.61T_m$)	NA	Boundary between Sn-rich dendrite and eutectic	Along Cu_6Sn_5 interface	Interfaces in Ag3Sn+Sn eutectic
				DBT
-40 °C ($0.47T_m$)	Deformation through Sn-rich dendrite	Boundary between Sn-rich dendrite and eutectic	NA	Interfaces in Ag3Sn+Sn eutectic
		DBT		
-85 °C ($0.38T_m$)	NA	Boundary between Sn-rich dendrite and eutectic	Along Cu_6Sn_5 interface	Interfaces in Ag3Sn+Sn eutectic
-120 °C ($0.31T_m$)	NA	Interfaces in Ag3Sn+Sn eutectic	Boundary between Sn-rich dendrite and eutectic	NA
-145 °C ($0.26T_m$)	NA	Interfaces in Ag3Sn+Sn eutectic	Boundary between Sn-rich dendrite and eutectic	Interfaces in Ag3Sn+Sn eutectic

the eutectic. However, at lower temperatures, specifically -120°C and -145°C, the deformation mechanism changes. Below -85°C, deformation primarily progresses along the interfaces within the Ag_3Sn + Sn eutectic (Table 4). In group 2b, deformation at -85°C and higher temperatures progresses along the Cu_6Sn_5 intermetallic interface, but specifically -120°C and -145°C, the deformation mechanism changes. Below -85°C, deformation primarily progresses along the boundary between Sn-rich dendrite and eutectic region (Table 4). In group 3, deformation during the micropillar compression experiment progresses along the interfaces within the Ag_3Sn + Sn eutectic across all temperatures (Table 4).

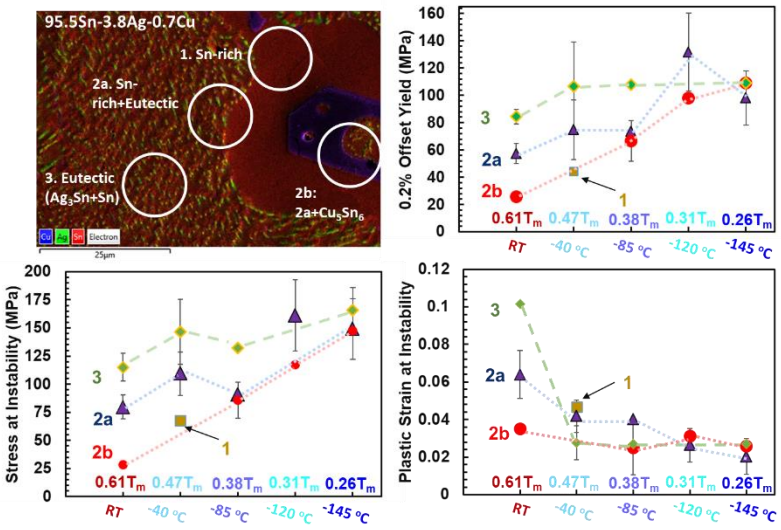


Figure 21. (a) Three pillar group categories, (b) 0.2% offset yield strength, (c) stress at instability, and (d) plastic strain at instability. The sudden drop in the plastic strain at instability in group 3 is probably due to the high Ag_3Sn fraction increasing stiffness and brittleness at cryogenic temperatures, localizing deformation at phase boundaries, and accelerating the ductile-to-brittle transition.

The 0.2% offset yield stress, stress at instability, and plastic strain at instability were further categorized based on pillar groups (Fig. 21). Each group exhibited an increase in stress at instability and 0.2% offset yield stress, along with a decrease in plastic strain at instability at cryogenic temperatures.

- Group 3: The 0.2% offset yield stress increased from 84 MPa at RT to 109 MPa at -145°C, while stress at instability rose from 115 MPa to 166 MPa. Plastic strain dropped sharply from 0.102 at RT to 0.027 at -145°C, with a sudden decrease from RT to -40°C followed by saturation.
- Group 2a: The 0.2% offset yield stress increased from 57 MPa at RT to 98 MPa at -145°C, while stress at instability rose from 80 MPa to 149 MPa. Plastic strain gradually decreased from 0.064 at RT to 0.02 at -145°C.
- Group 2b: The 0.2% offset yield stress increased from 26 MPa at RT to 109 MPa at -145°C, while stress at instability rose from 28 MPa to 148 MPa. However, plastic strain at instability in this group showed no significant change at cryogenic temperatures.
- Since group 1 is only present at -40°C, a direct trend across all temperatures is not observed.

Group 3 (Ag_3Sn + Sn eutectic) demonstrated the highest stress at instability and 0.2% offset yield stress due to its composite nature and higher fraction of the hard Ag_3Sn intermetallic phase (Fig. 21b, c). The following conclusions can be made from these results:

- In Group 2a, plastic strain gradually decreases with decreasing temperature. Unlike Group 3, which exhibits a sudden drop, Group 2a has a more gradual decrease due to the balance between the Sn-rich phase and eutectic regions, allowing for some residual plasticity.
- Group 2b consistently has the lowest plastic strain across all temperatures. This is probably caused by the presence of Cu_6Sn_5 intermetallic compounds, which increase local stiffness and restrict plastic deformation. Cu_6Sn_5 is a hard and brittle phase that limits dislocation movement, making the material prone to early instability. The lack of significant change in strain across temperatures suggests that the material is already in a constrained deformation regime, dominated by intermetallic interfaces rather than ductile phase behavior.
- Group 3 shows a sharp decrease in the plastic strain at instability as the temperature drops below room temperature (RT). This is probably caused by the high fraction of the Ag_3Sn intermetallic phase that significantly influences the mechanical response. At cryogenic temperatures, Ag_3Sn is much stiffer and more brittle than the Sn-rich phase. As temperature decreases, deformation becomes increasingly localized at phase boundaries, reducing plasticity and accelerating the ductile-to-brittle transition.

This analysis highlights the importance of understanding localized deformation in solder joints under cryogenic conditions, which is crucial for in-space applications. Depending on the soldering environment (microgravity vs. gravity), solder composition, soldering process parameters, and solder microstructure, the solder behavior can vary under cryogenic temperatures. Such as, The absence of natural convection and reduced fluid motion in microgravity results in a coarser grain structure compared to terrestrial conditions, leading to fewer nucleation sites for the β-Sn to α-Sn transformation and slower phase transformation kinetics. However, the coarser grain structure in microgravity is expected to accelerate the ductile-to-brittle transition due to faster transformation kinetics. To mitigate these effects, the addition of trace alloying elements such as indium, bismuth, and silver can help prevent the β-to-α Sn transformation, while rapid cooling can refine the grain structure, improving mechanical stability. Implementing both strategies together is essential to suppress both the phase transformation and the ductile-to-brittle transition, ensuring improved solder reliability in microgravity environment.

Summary:

- This study investigated the 40Pb-60Sn solder sample prepared during in-space soldering investigation in microgravity conditions onboard ISS and terrestrial conditions at NASA Marshall, along with solder samples prepared in Dr. Pathak's lab at Iowa State University Ames (ISU). This study also extended to non-ISSI solders such as off-eutectic 50Pb-50Sn and 95.5Sn-3.8Ag-0.7Cu solders prepared in terrestrial conditions at ISU.
- The effects of microgravity on solder joint formation, microstructure, and mechanical behavior was investigated using advanced characterization techniques. Non-destructive 3D tomography revealed that larger voids occurred only in microgravity due to the absence of natural convection. A quantitative framework demonstrated that ISSI microgravity 40Pb-60Sn wire wrap solder had ~13× more voids than ISSI terrestrial 40Pb-60Sn solder, while ISSI 40Pb-60Sn wire feed solder had 4× more voids than ISSI 40Pb-60Sn wire wrap solder.
- The study examined Marangoni convection in microgravity, which induced Pb-rich dendrite accumulation in hot regions, lowering hardness (0.11 GPa) at hot sites compared to colder sites (0.15 GPa). Aging studies on terrestrial 40Pb-60Sn and terrestrial 50Pb-50Sn solder showed grain coarsening and hardness reduction, affecting long-term reliability.

- Cryogenic investigations using in-situ SEM micropillar compression a ductile-to-brittle transition (DBT) in solder. In 40Pb-60Sn, strength increased from 45.9 MPa at RT to 161.4 MPa at -145°C, contradicting the expected β-to-α phase transformation. Additionally, XRD analysis showed no phase transformation. Furthermore, plastic strain dropped from 0.109 at RT to 0.062 at -85°C, saturating below this temperature, with deformation localized at Pb-Sn phase boundaries, confirming DBT.
- Similarly, in ISU terrestrial 95.5Sn-3.8Ag-0.7Cu solder, plastic strain decreased from 0.06 at RT to 0.04 at -40°C, stabilizing at 0.03 below -40°C, which indicated the effect of ductile to brittle transition. Strength increased significantly, with instability stress rising from 78.1 MPa at RT to 153 MPa at -145°C.
- All pillar groups in ISU terrestrial 95.5Sn-3.8Ag-0.7Cu solder showed an increase in stress at instability and 0.2% offset yield stress with decreasing temperature, while plastic strain at instability decreased. Group 3 (Ag_3Sn + Sn eutectic) exhibited the highest strength but experienced a sharp drop in plastic strain below room temperature, whereas Group 2a showed a more gradual decline. Group 2b had the lowest plastic strain across all temperatures due to the presence of Cu_6Sn_5 intermetallic compounds, which restricted deformation.
- The high fraction of Ag_3Sn in Group 3 made it stiffer and more brittle at cryogenic temperatures, leading to localized deformation at phase boundaries and a rapid ductile-to-brittle transition. Group 2b, containing Cu_6Sn_5, exhibited a minimal change in plastic strain, indicating a constrained deformation regime. In contrast, Group 2a, with a mix of Sn-rich and eutectic phases, retained some plasticity, resulting in a more gradual transition.
- This work on solder behavior at cryogenic temperature establishes a baseline for solder behavior across extreme temperatures and providing insights for future microgravity metal joining experiments on the ISS or other reduced gravity platforms.

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